還原及還原SAP HANA資料庫

Snap Creator Framework

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還原及還原SAP HANA資料庫

您可以使用SAP HANA Studio和Snap Creator來還原及還原SAP HANA資料庫。

1. 在SAP HANA Studio中：
   a. 針對SAP HANA系統選取「恢復」。
   b. SAP HANA系統已關機。
   c. 選取恢復類型。
   d. 提供記錄備份位置。
   e. 此時會顯示資料備份清單
   f. 選取備份以查看外部備份ID。
2. 若儲存系統只執行叢集Data ONTAP 式的不適用功能：
   a. 只有在使用非最新的備份進行還原時才需要。
   b. 僅SnapRestore 適用於主儲存設備的「Volume」。
   c. 停用SnapVault 相依關係
3. 在Snap Creator中：
   a. 針對SAP HANA系統選取「還原」。
   b. 視主要儲存設備的備份可用度而定、選取從主要或次要儲存設備還原。
   c. 選取儲存控制器、磁碟區名稱和Snapshot複本名稱。Snapshot複本名稱與先前的備份ID有關。
   d. 對於多節點SAP HANA系統、需要還原多個磁碟區：
      i. 選擇*新增更多還原項目*。
      ii. 選取儲存控制器、磁碟區名稱和Snapshot複本名稱。
      iii. 針對所有必要的磁碟區重複此程序。
   e. 對於多租戶資料庫容器（MDC）單一租戶資料庫系統、系統和租戶資料庫都會還原。
   f. 還原程序已開始
   g. 所有磁碟區的還原均已完成。
4. 在資料庫節點上、卸載並掛載所有資料磁碟區、以清除「過時的NFS處理代碼」。
5. 在SAP HANA Studio中：
   a. 在備份清單中選取* Refresh*。
   b. 選取可用的備份以進行恢復（綠色項目）。
   c. 開始恢復程序。
   d. 對於多租戶資料庫容器（MDC）單租戶資料庫系統、請先針對系統資料庫、然後針對租戶資料庫開始還原程序。
   e. SAP HANA系統隨即啟動。
6. （選用）恢復SnapVault 所有已還原磁碟區的彼此之間的關係。

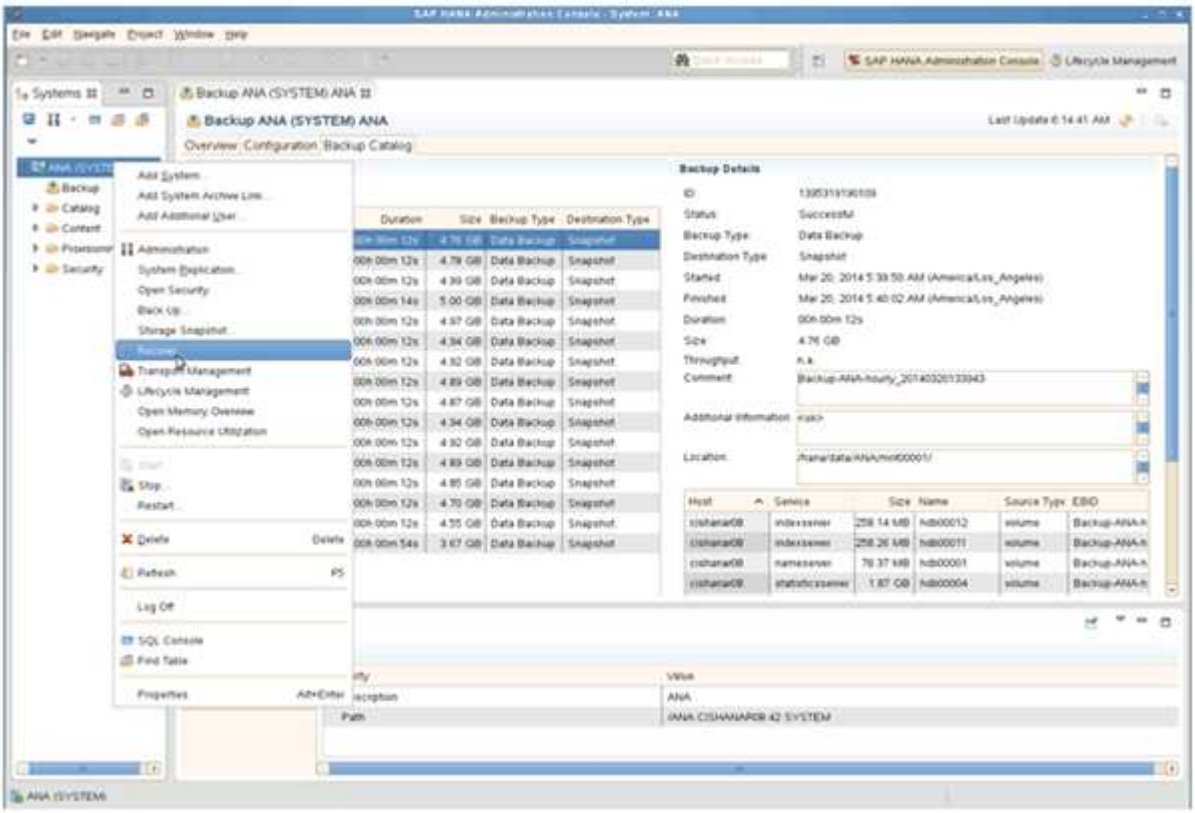
在儲存系統中、只有在使用非最新備份進行還原時、才需要執行此步驟。

從主要儲存設備還原及還原資料庫

您可以從主要儲存設備還原及還原資料庫。

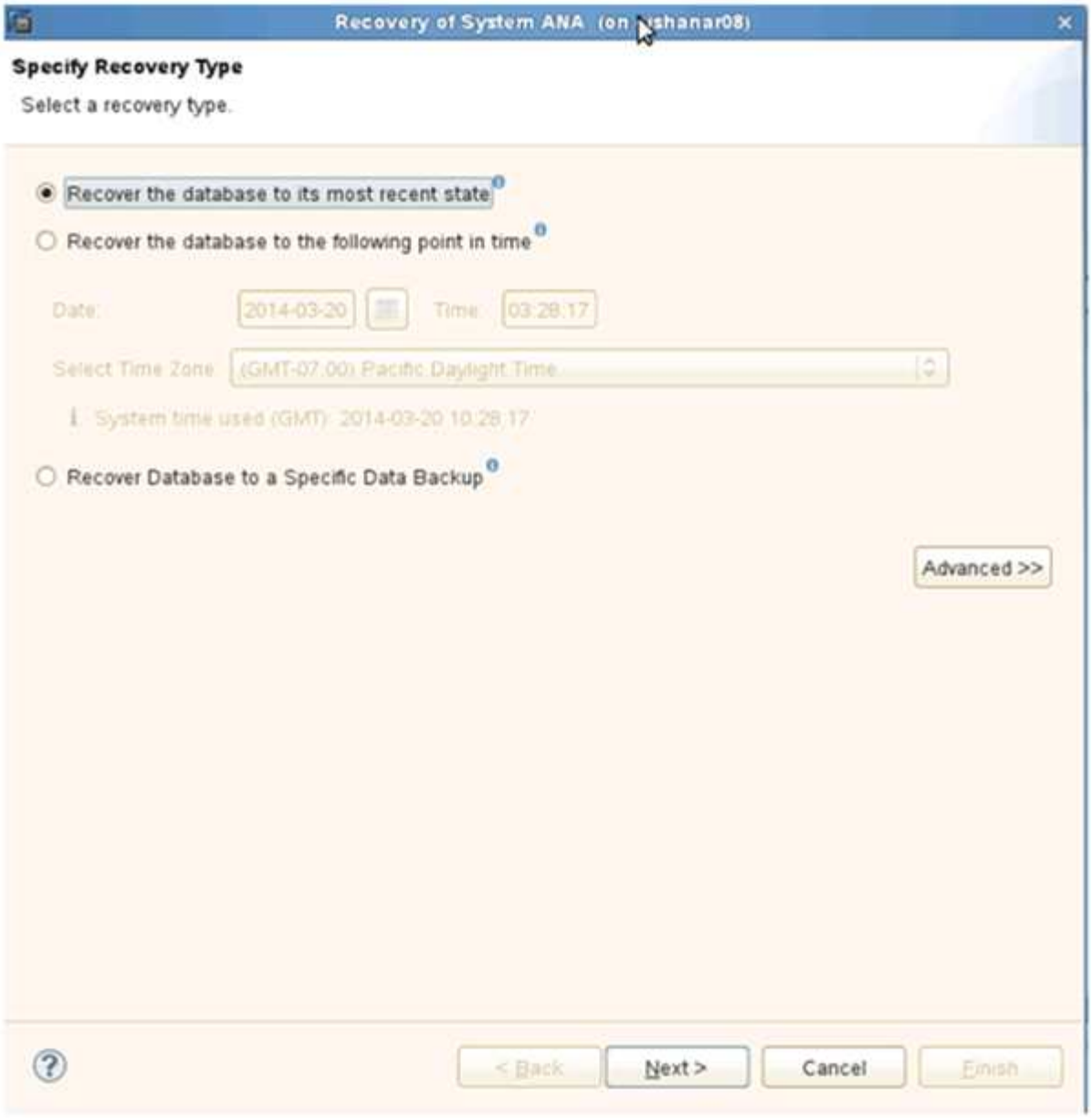
您無法從Snap Creator還原檔案型備份複本。

1. 在SAP HANA Studio中、針對SAP HANA系統選取* recover*。

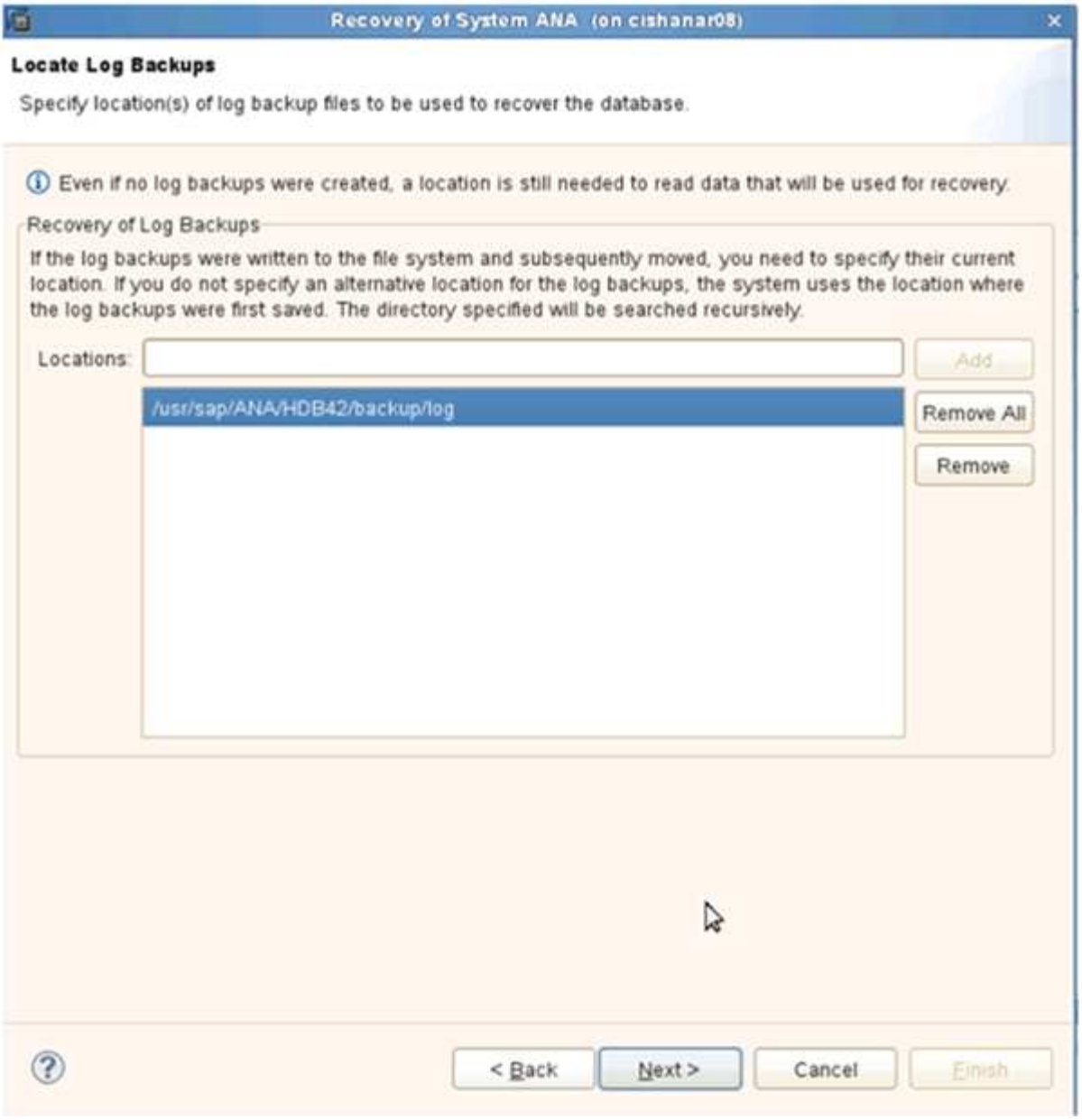


SAP HANA系統隨即關機。

2. 選取恢復類型、然後按一下*「下一步」*。

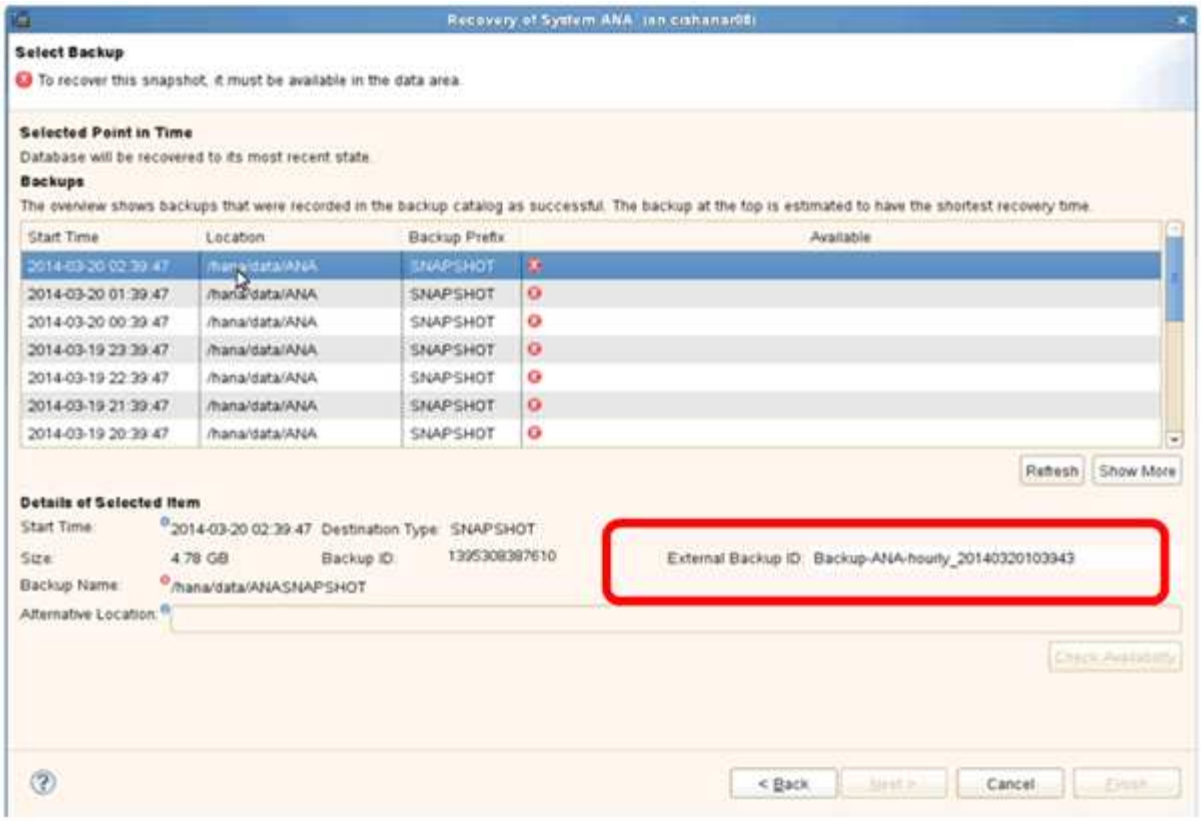


3. 提供記錄備份位置、然後按一下*下一步*。



您所看到的可用備份清單是根據備份目錄的內容而定。

4. 選取所需的備份並記錄外部備份ID。



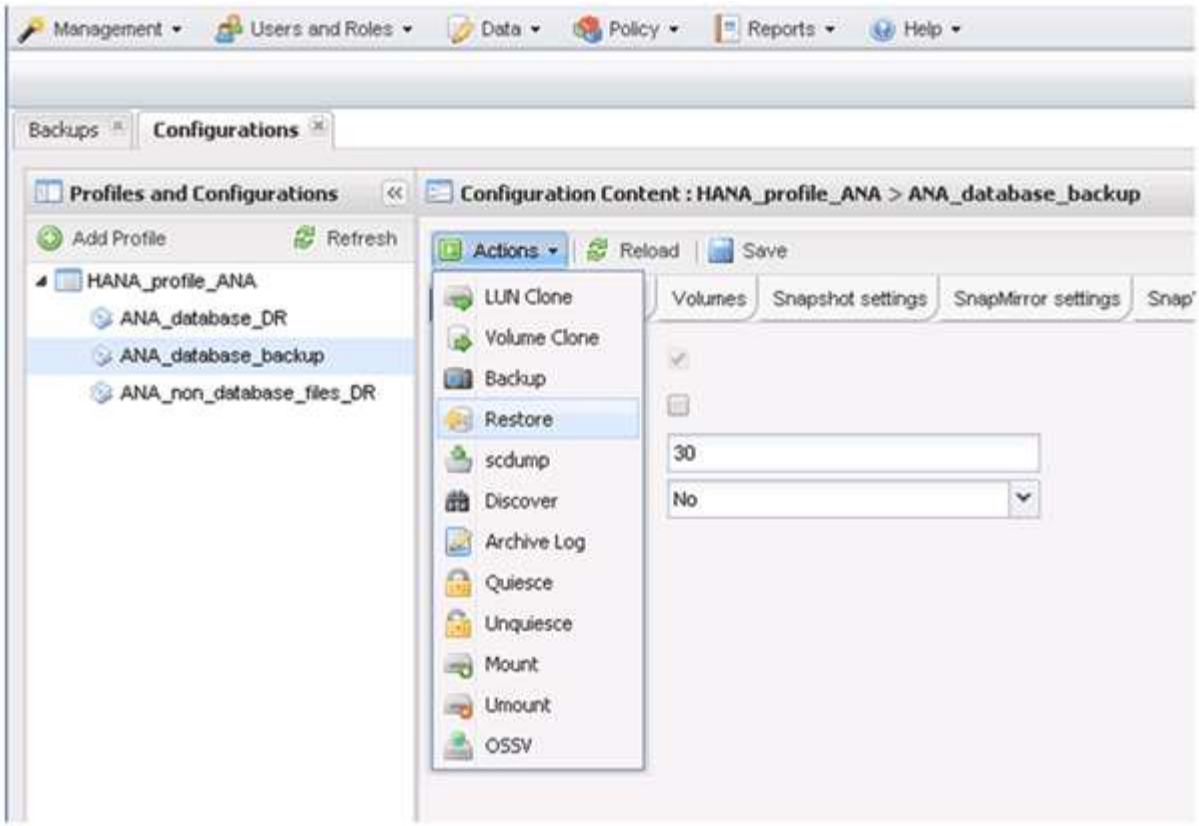
5. 停用SnapVault 「不合作關係」。

> 只有叢集Data ONTAP 式的不需要執行此步驟。

如果您需要還原的Snapshot複本比目前用做SnapVault 為基礎Snapshot複本的Snapshot複本舊、您必須先停用SnapVault 叢集Data ONTAP 式故障恢復中的不再使用該功能。若要這麼做、請在備份叢集主控台上執行下列命令：

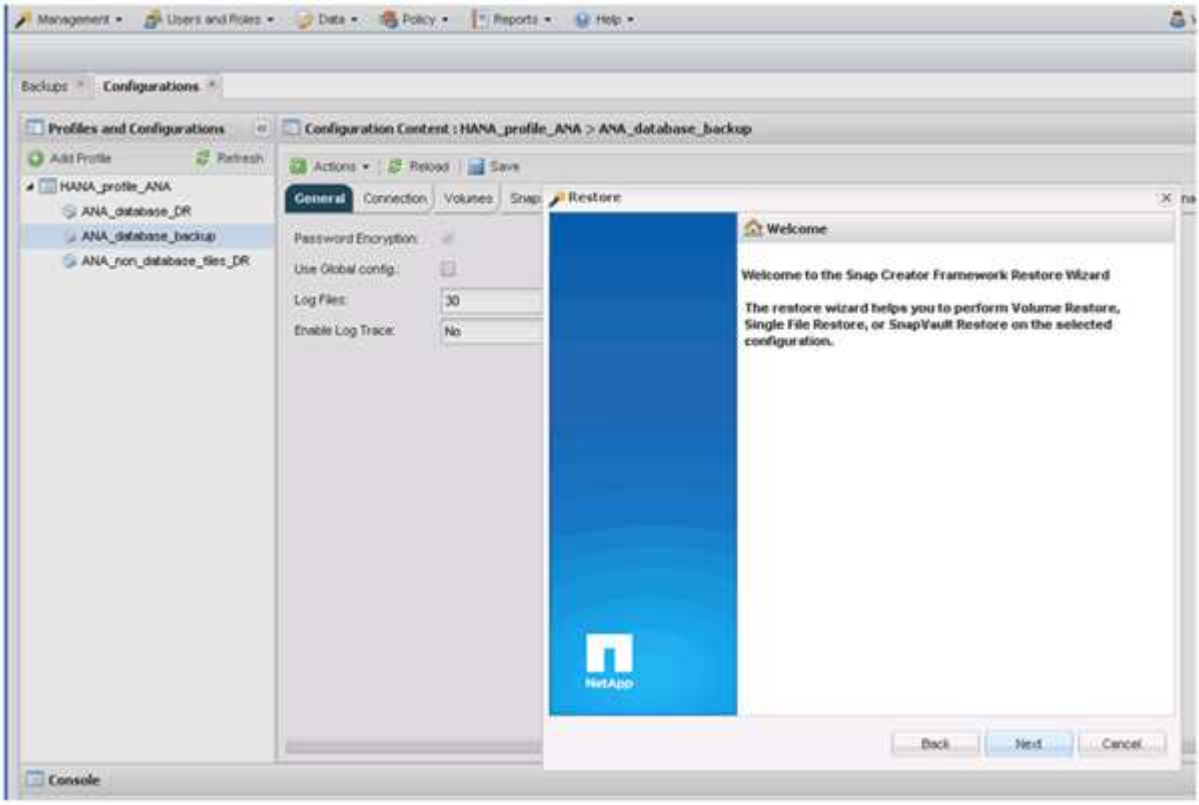
```
hana::> snapmirror quiesce -destination-path hana2b:backup_hana_data
Operation succeeded: snapmirror quiesce for destination
hana2b:backup_hana_data.

hana::> snapmirror delete -destination-path hana2b:backup_hana_data
Operation succeeded: snapmirror delete the relationship with destination
hana2b:backup_hana_data.

hana::> snapmirror release -destination-path hana2b:backup_hana_data
[Job 6551] Job succeeded: SnapMirror Release Succeeded
```

6. 在Snap Creator GUI中、選取SAP HANA系統、然後選取* Actions **>** Restore *。

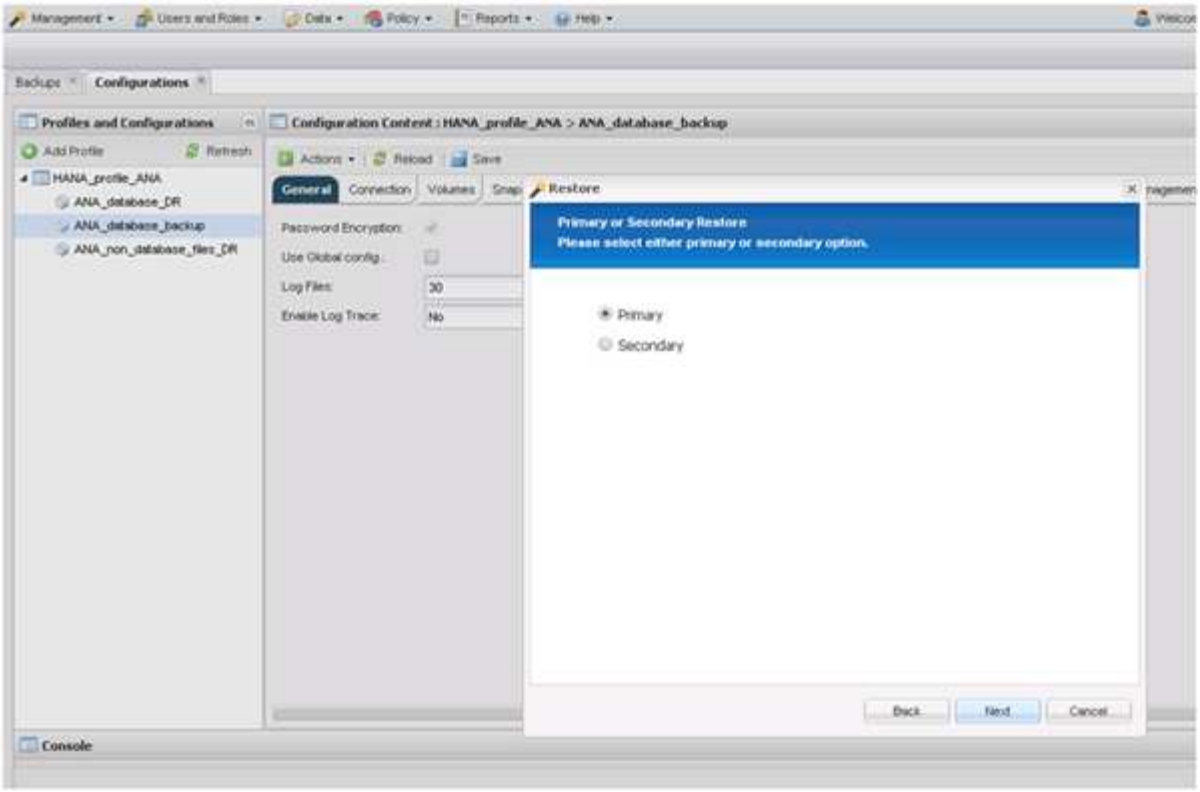


「歡迎使用SnapCreator Framework還原精靈」畫面隨即出現。

7. 單擊 * 下一步 * 。

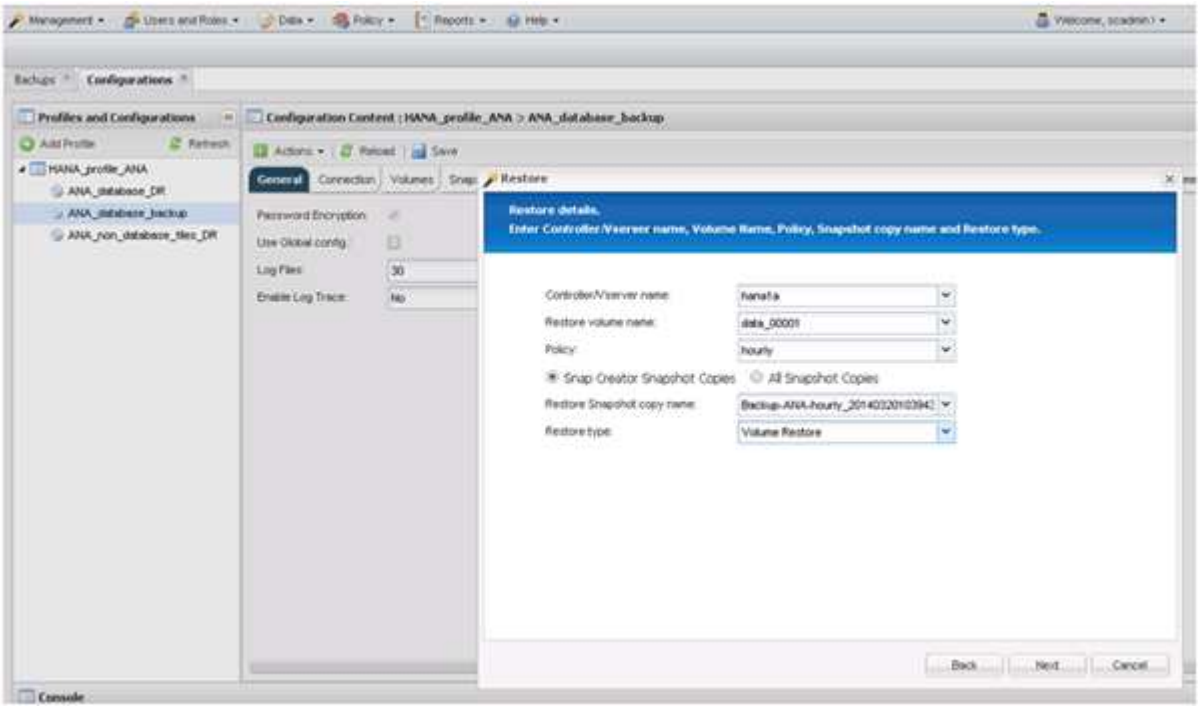


8. 選擇* Primary*（主*）、然後按* Next*（下一步*）。

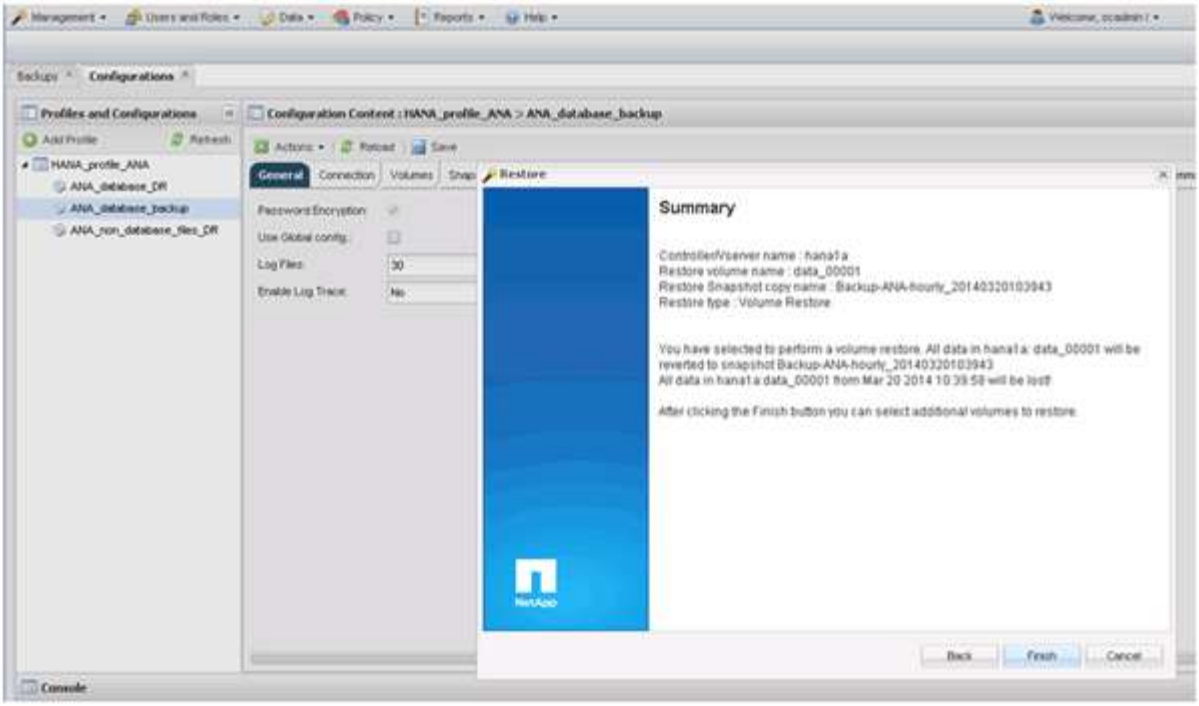


9. 選取「從主要儲存設備還原」。
10. 選取儲存控制器、磁碟區名稱和Snapshot名稱。

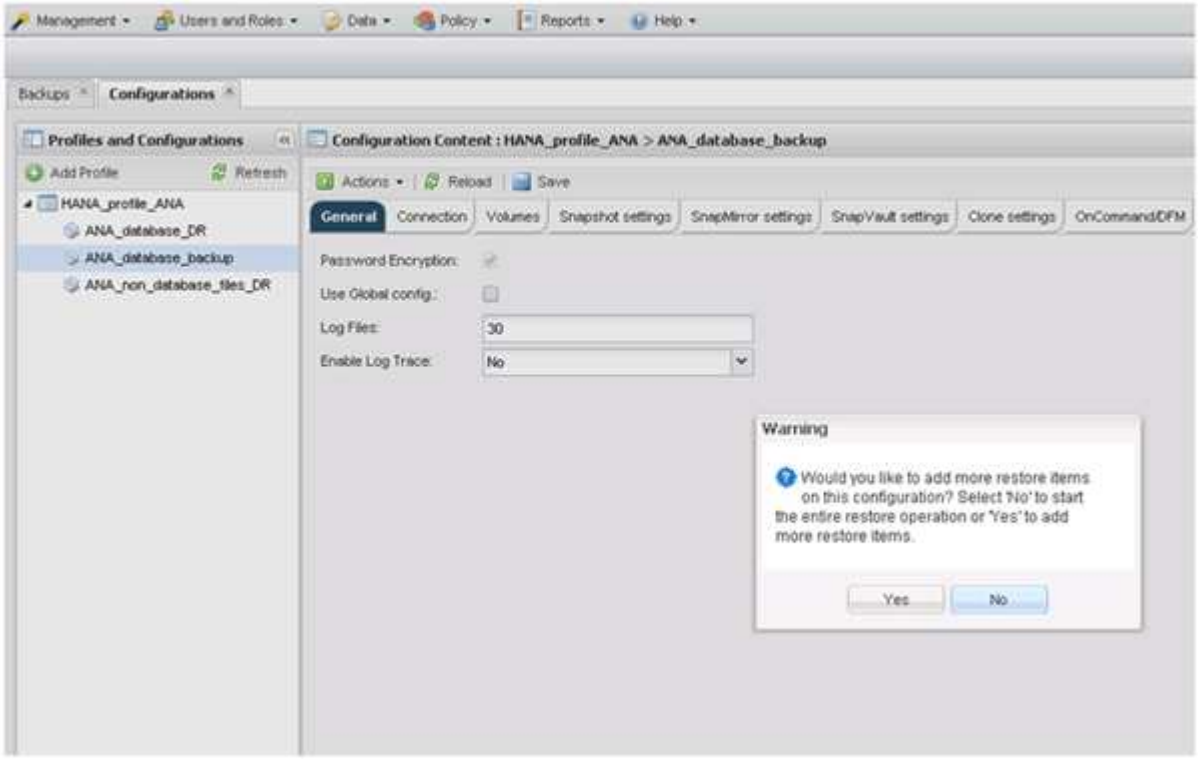
    Snapshot名稱與SAP HANA Studio中選取的備份ID有關。



11. 單擊*完成*。

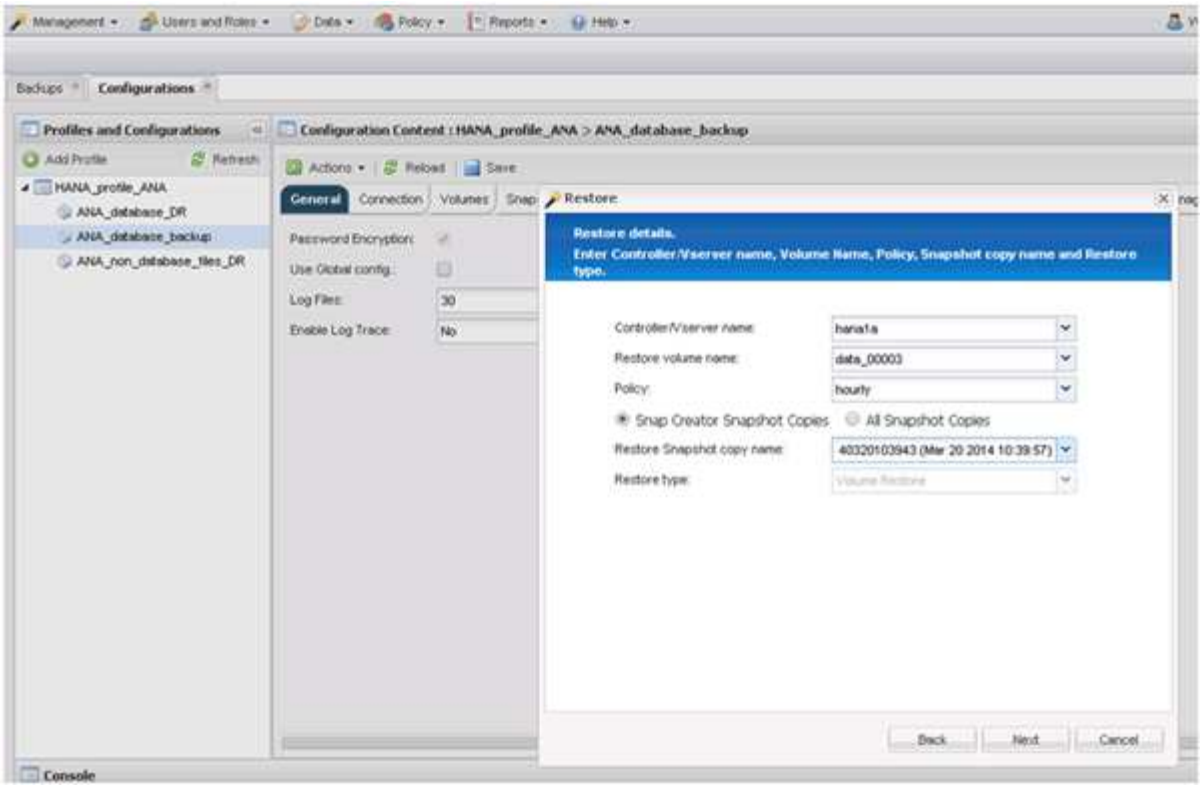


12. 按一下「是」以新增更多還原項目。

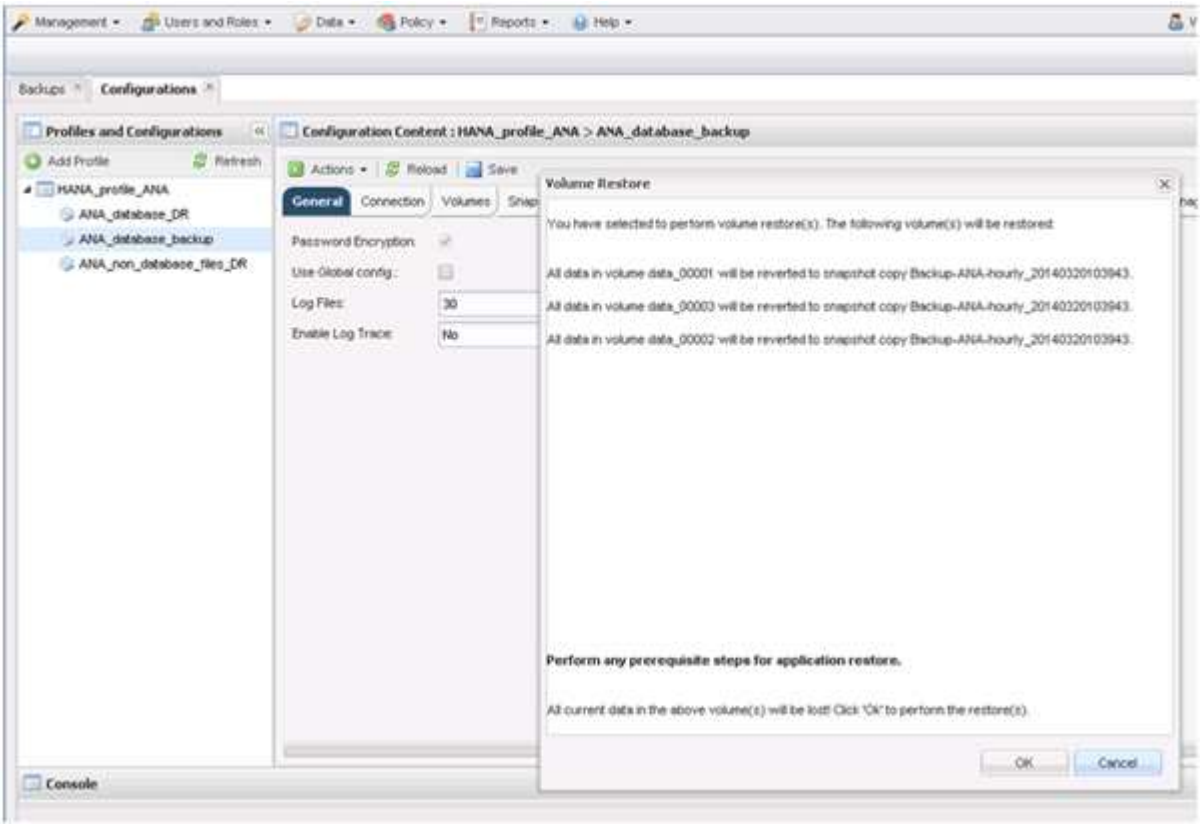


13. 選取儲存控制器、其他磁碟區名稱和Snapshot名稱。

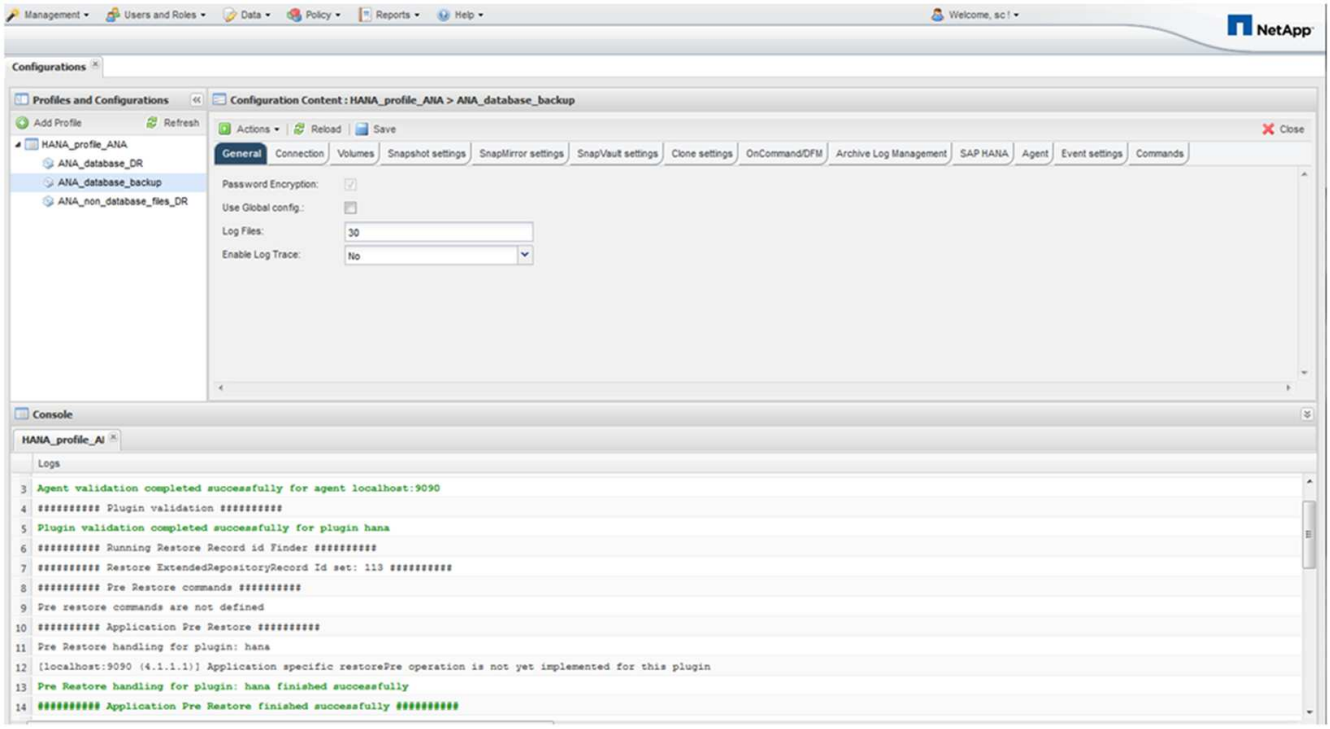
    Snapshot名稱與SAP HANA Studio中選取的備份ID有關。



14. 重複步驟10到13、直到新增所有必要的磁碟區為止；在我們的範例中、需要選取data_00001、data_00002和data_00003來進行還原程序。
15. 選取所有磁碟區後、按一下「確定」開始還原程序。



還原程序隨即開始。



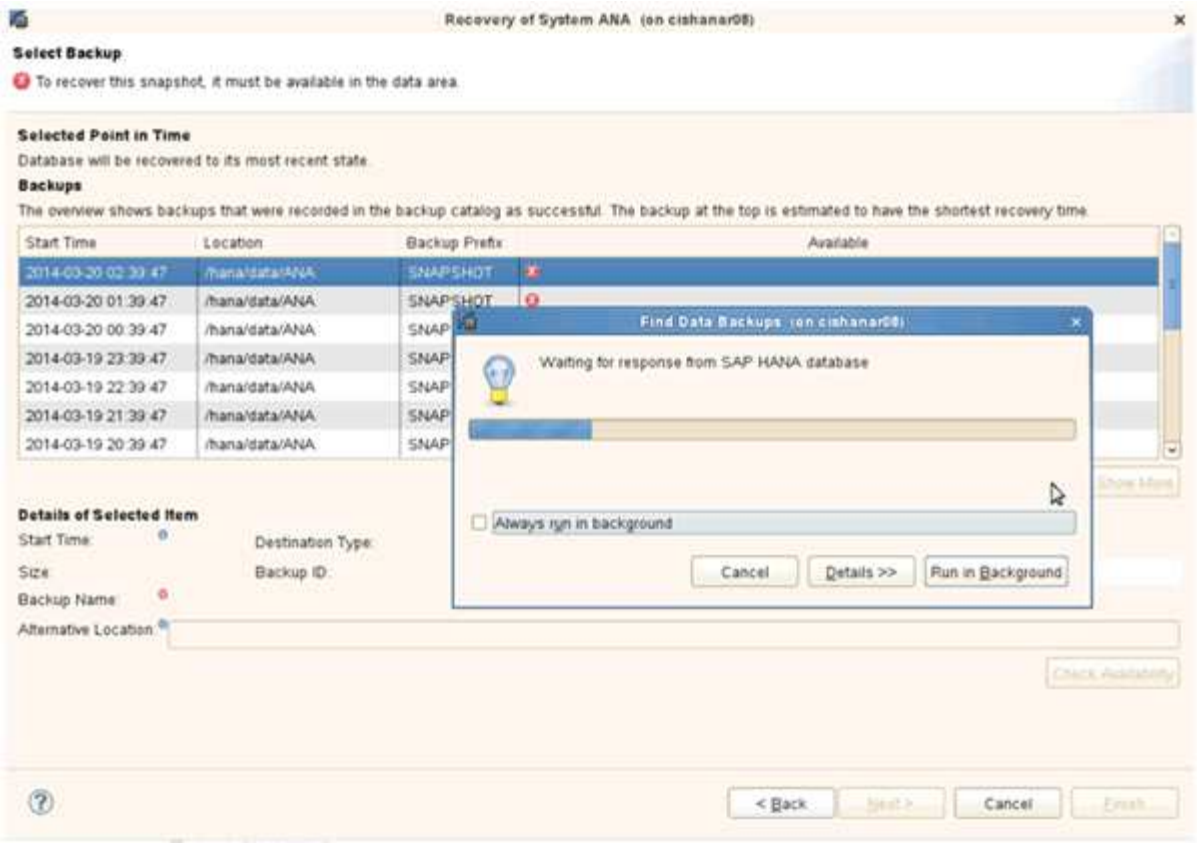
等待還原程序完成。

16. 在每個資料庫節點上、重新掛載所有資料磁碟區、以清除過時的NFS處理程序。

    在此範例中、所有三個磁碟區都需要在每個資料庫節點上重新掛載。

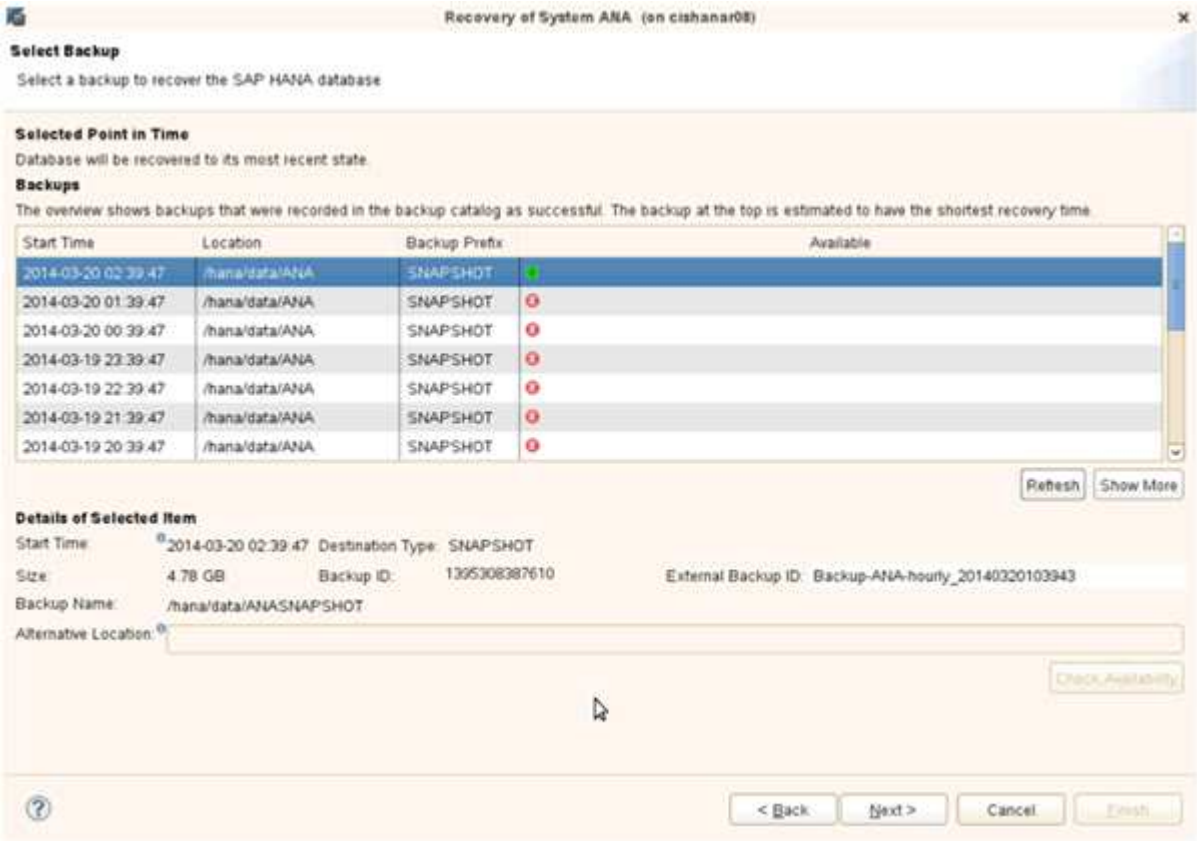
```
mount -o remount /hana/data/ANA/mnt00001
mount -o remount /hana/data/ANA/mnt00002
mount -o remount /hana/data/ANA/mnt00003
```

17. 前往SAP HANA Studio並按一下* Refresh*以更新可用備份清單。

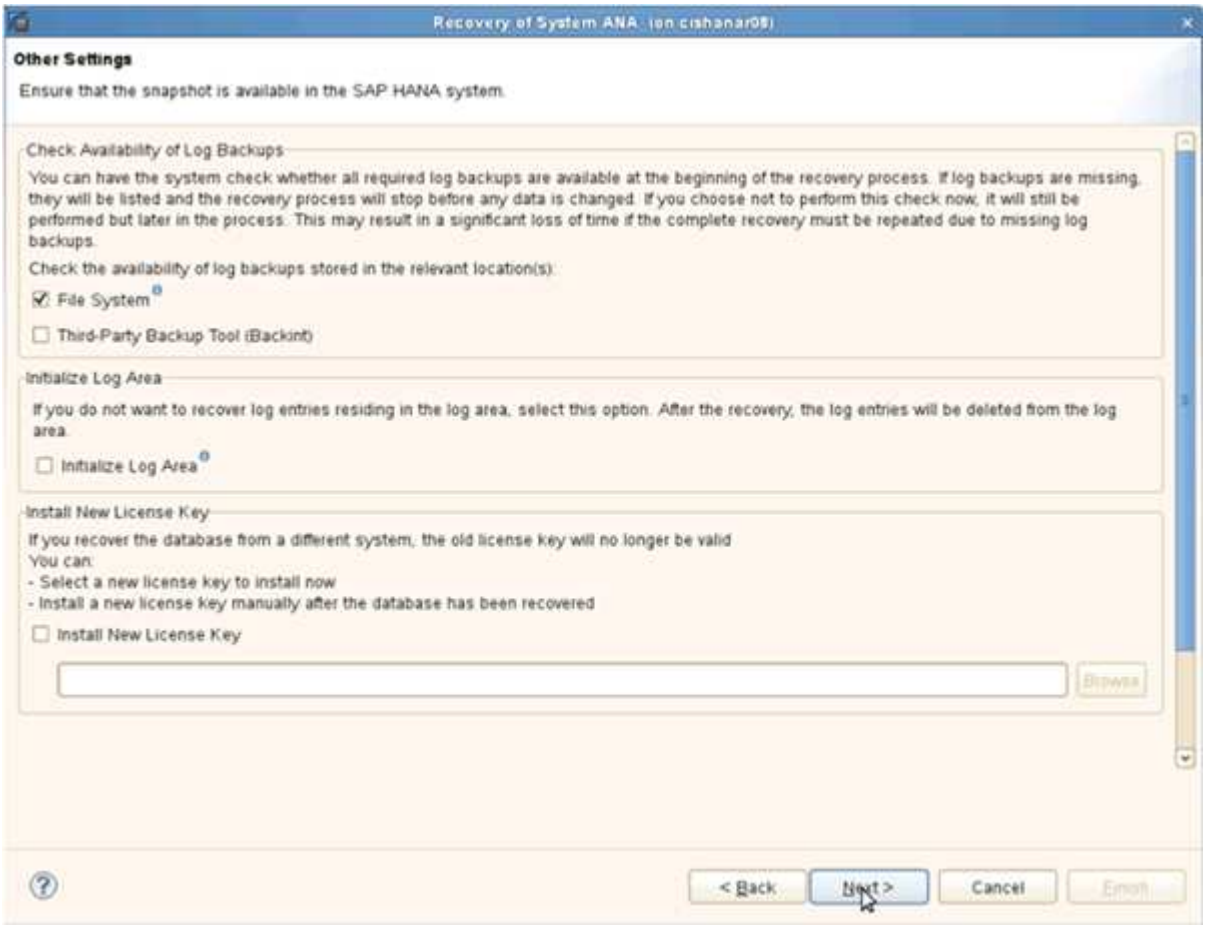


使用Snap Creator還原的備份會在備份清單中顯示綠色圖示。

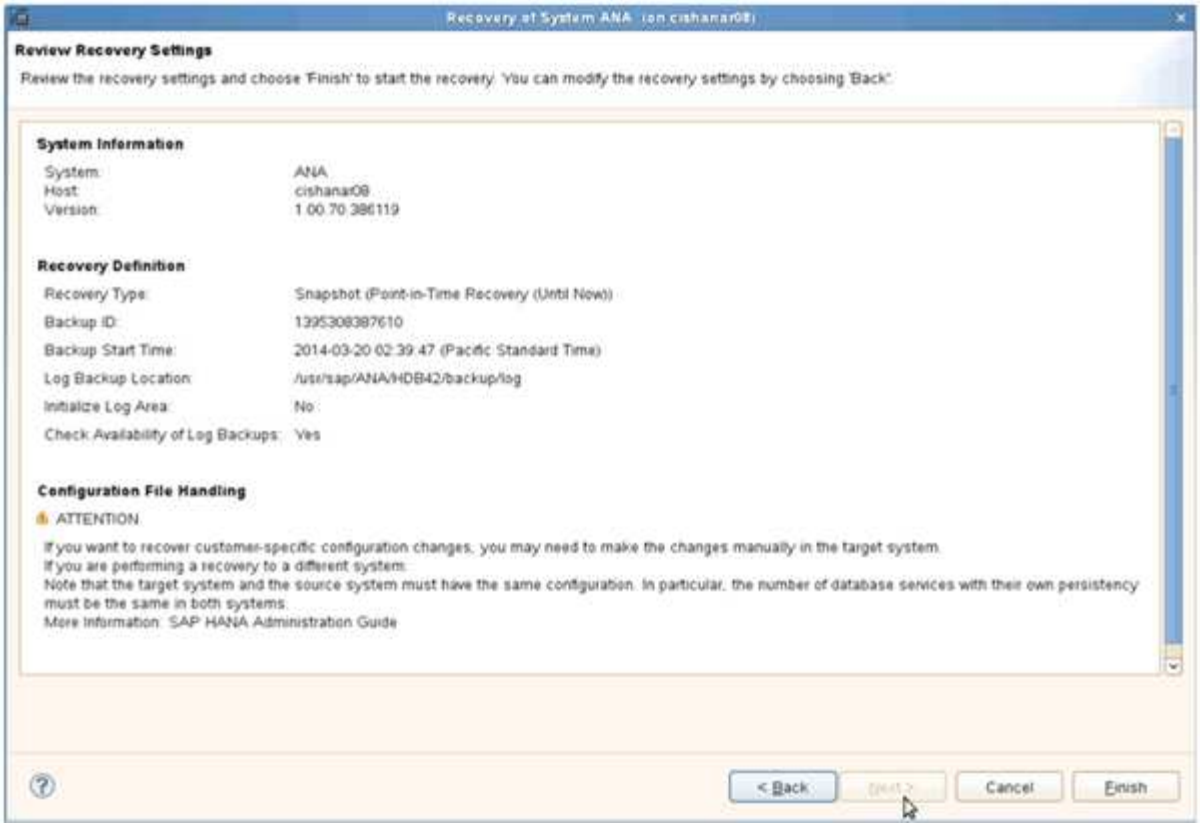
18. 選取備份、然後按一下*下一步*。



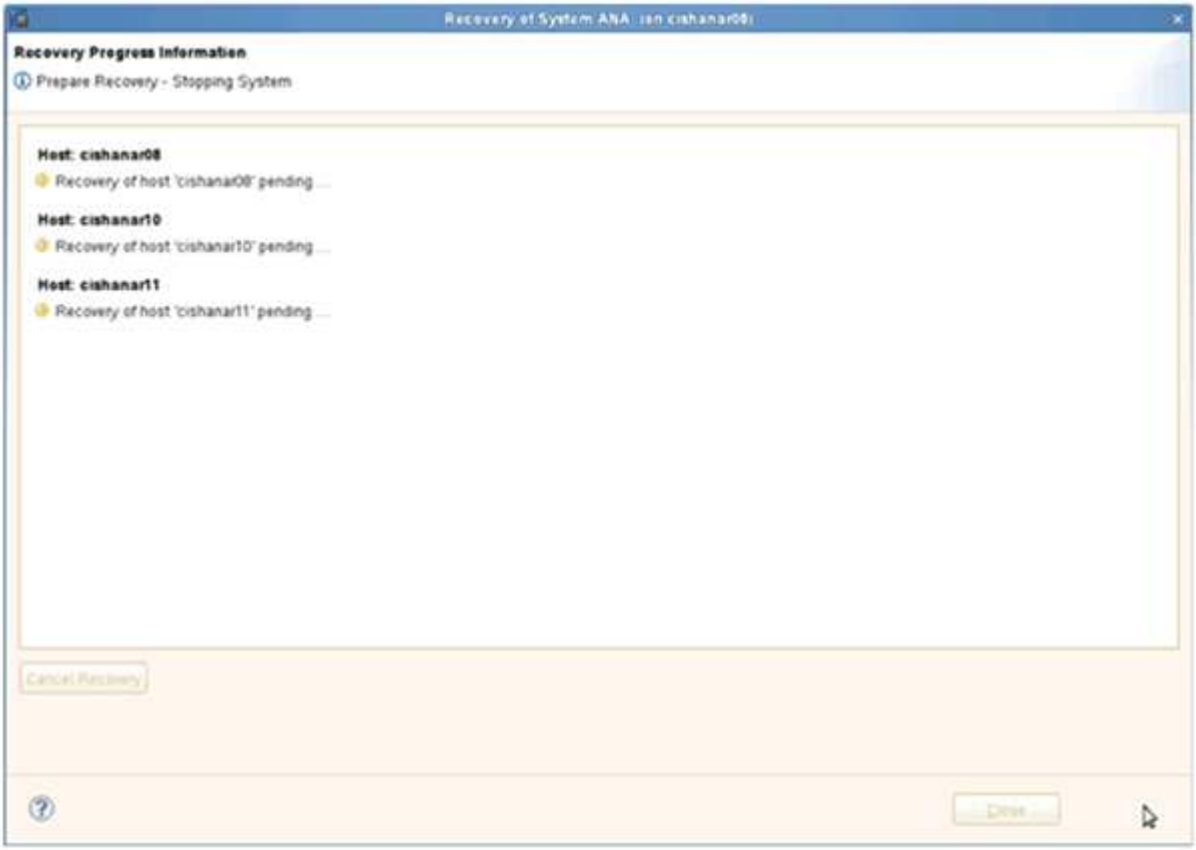
19. 視需要選取其他設定、然後按一下*下一步*。



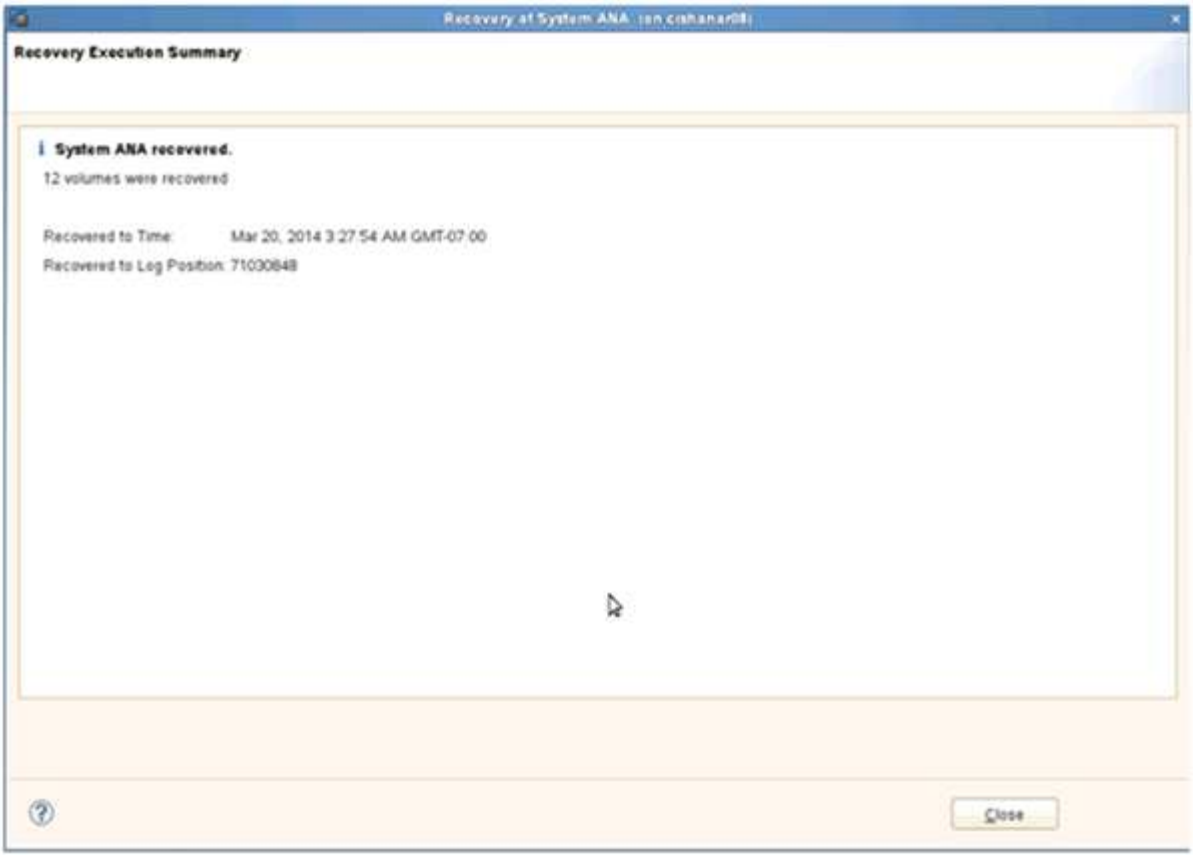
20. 單擊*完成*。



恢復程序隨即開始。



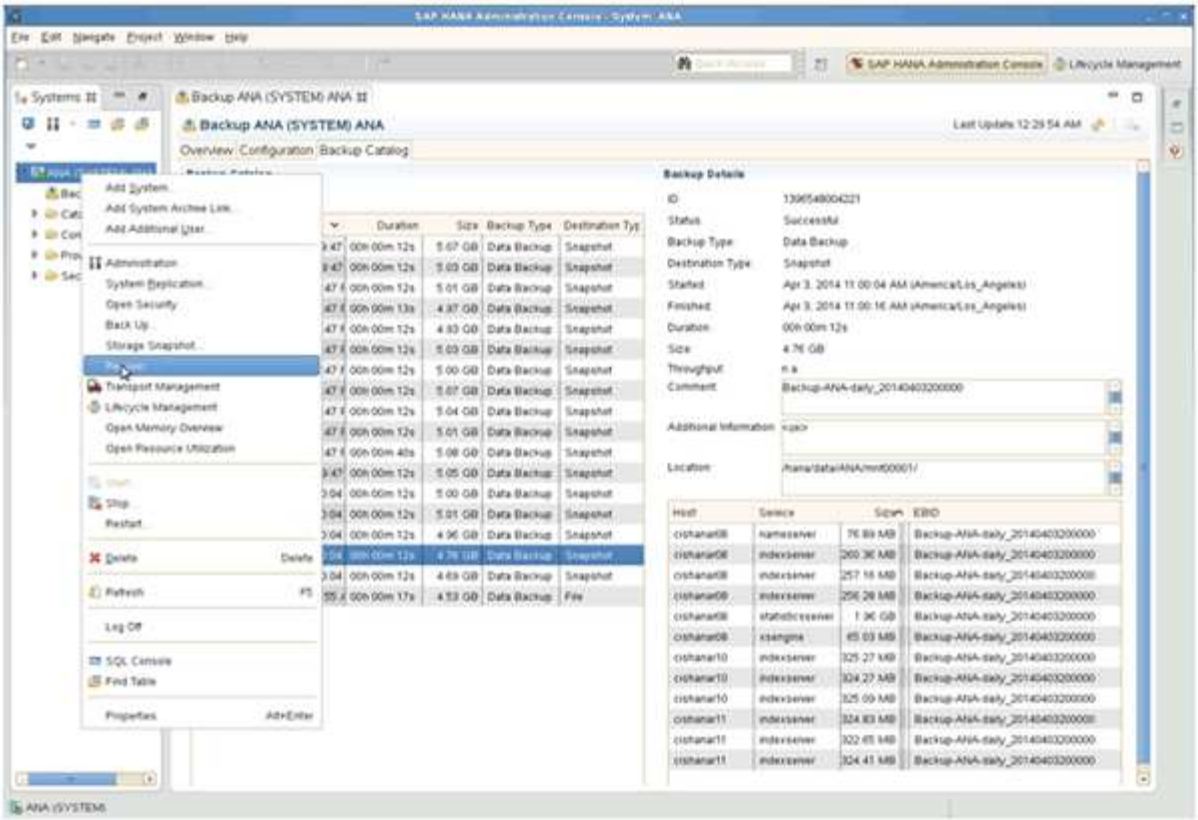
21. 恢復完成後、SnapVault 如有需要、請繼續恢復彼此之間的關係。



從二線儲存設備還原及還原資料庫

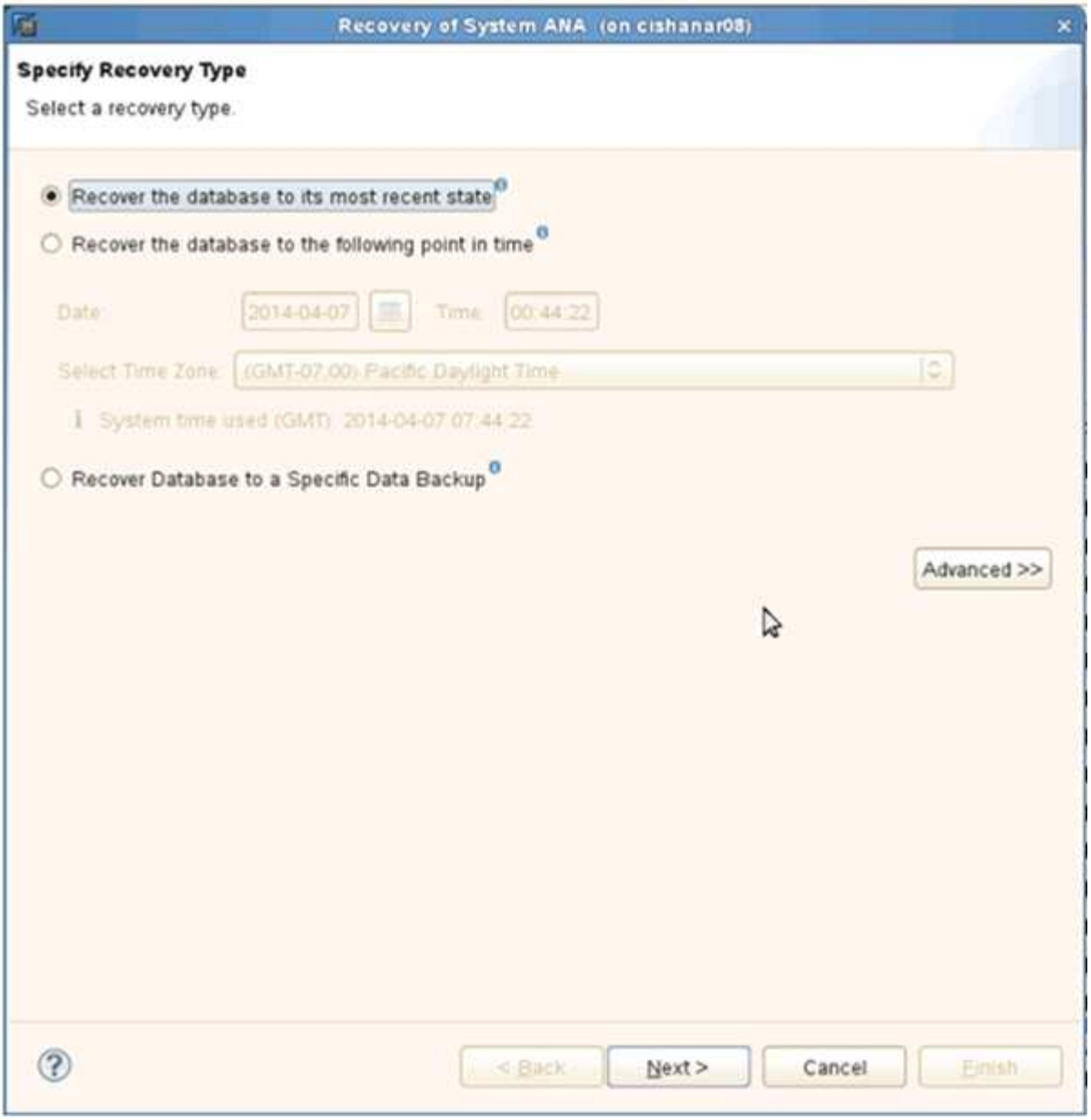
您可以從次要儲存設備還原及還原資料庫。

1. 在SAP HANA Studio中、針對SAP HANA系統選取* recover*。

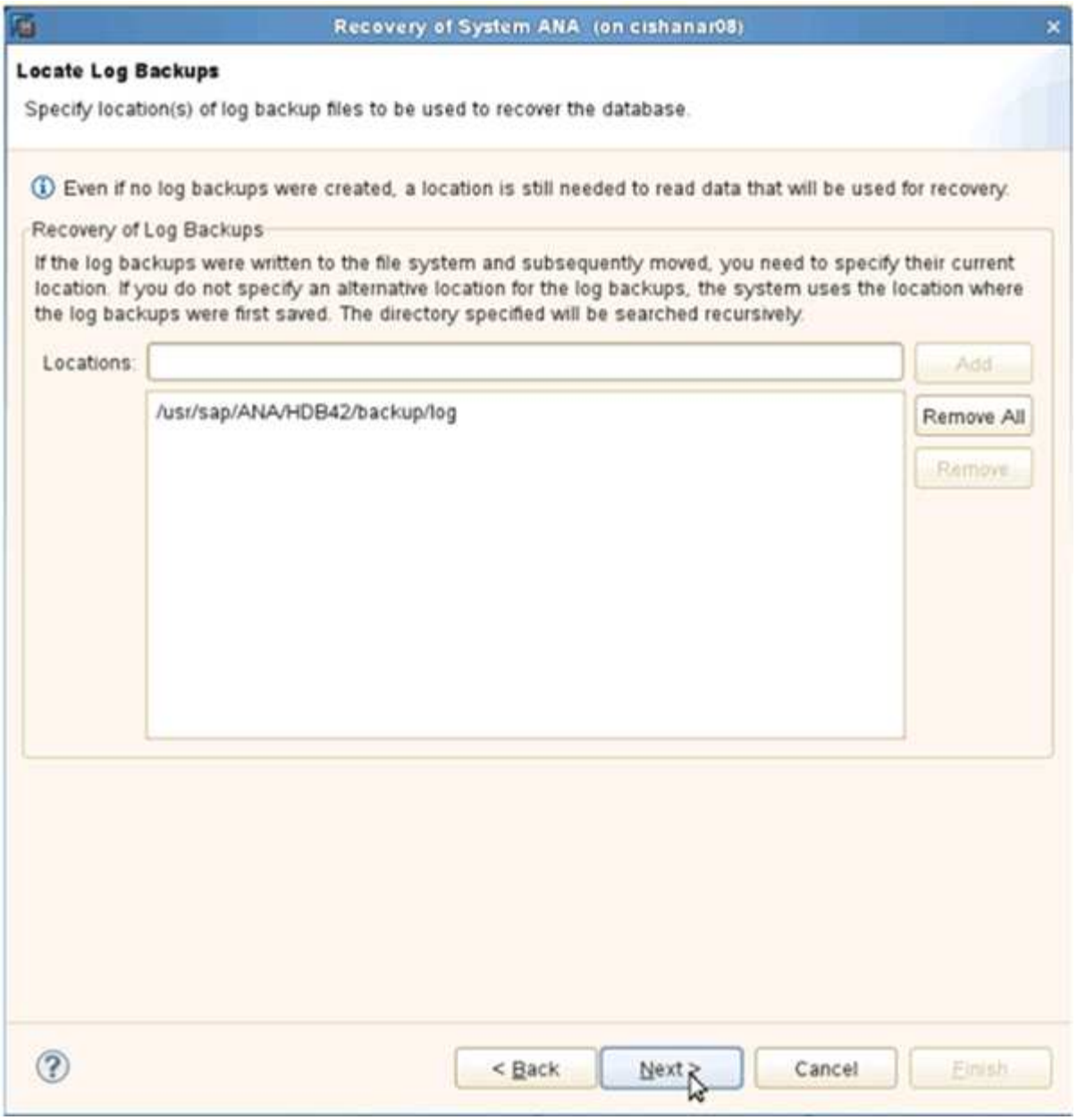


SAP HANA系統將會關閉。

2. 選取恢復類型、然後按一下*「下一步」*。

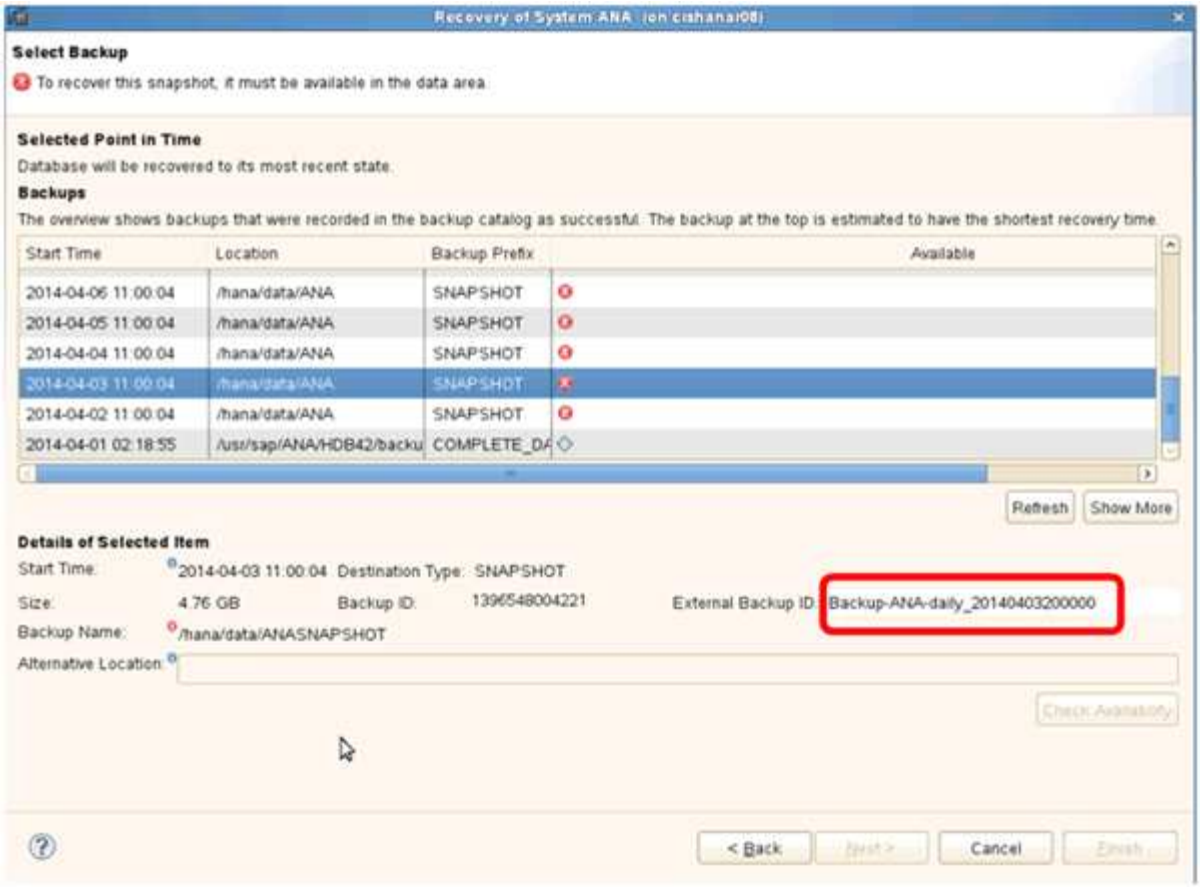


3. 提供記錄備份位置、然後按一下*下一步*。

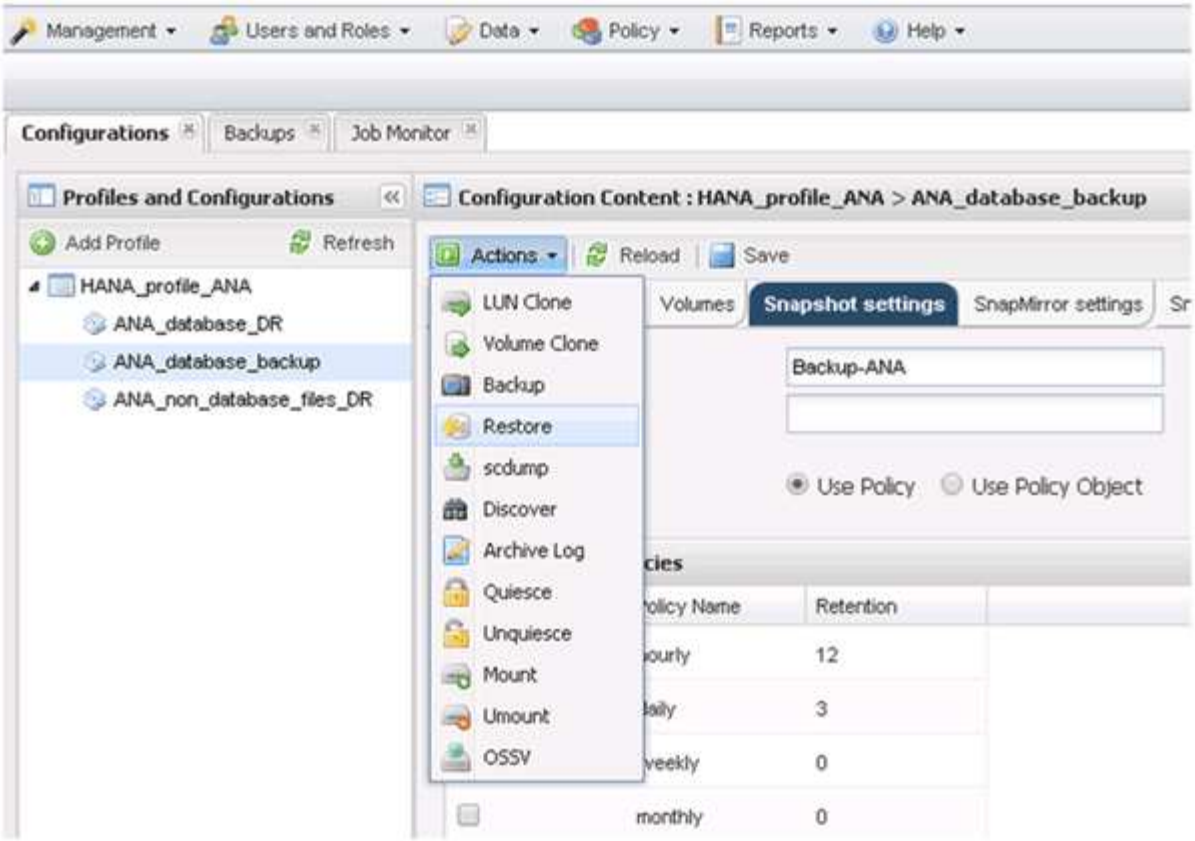


可用備份清單會根據備份目錄的內容顯示。

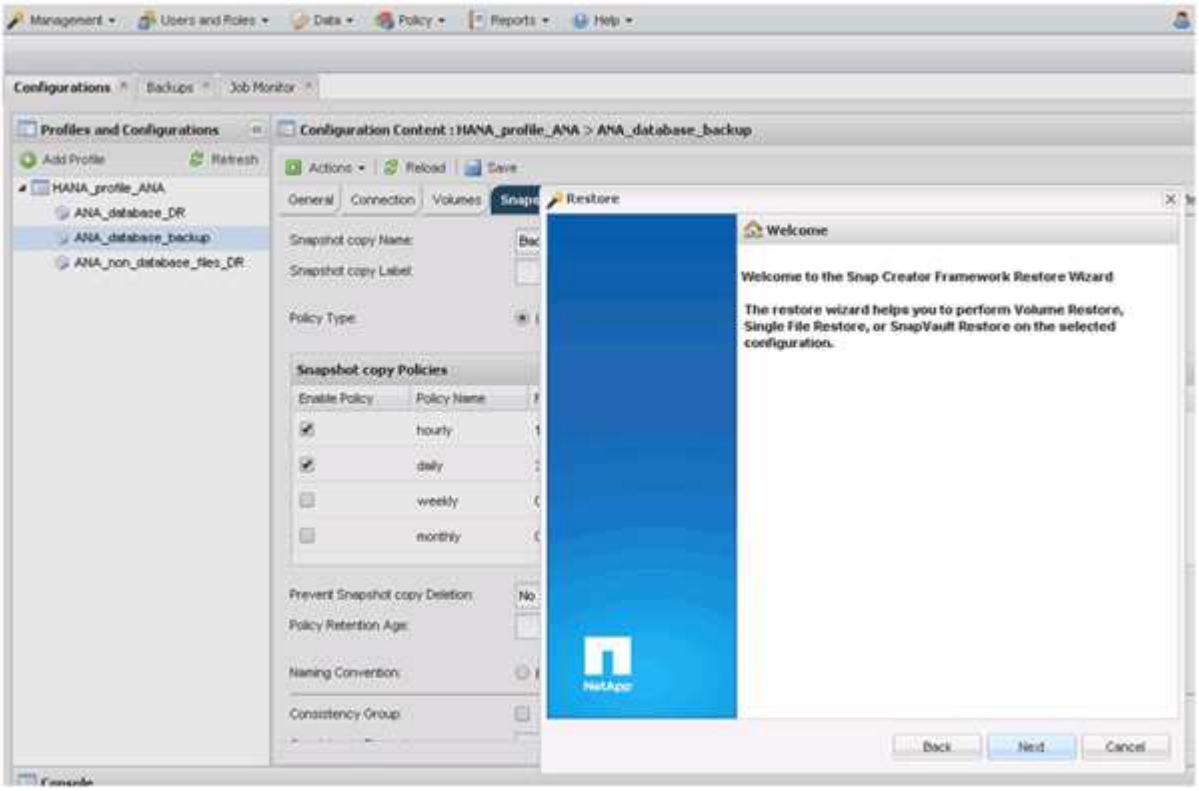
4. 選取所需的備份並記下外部備份ID。



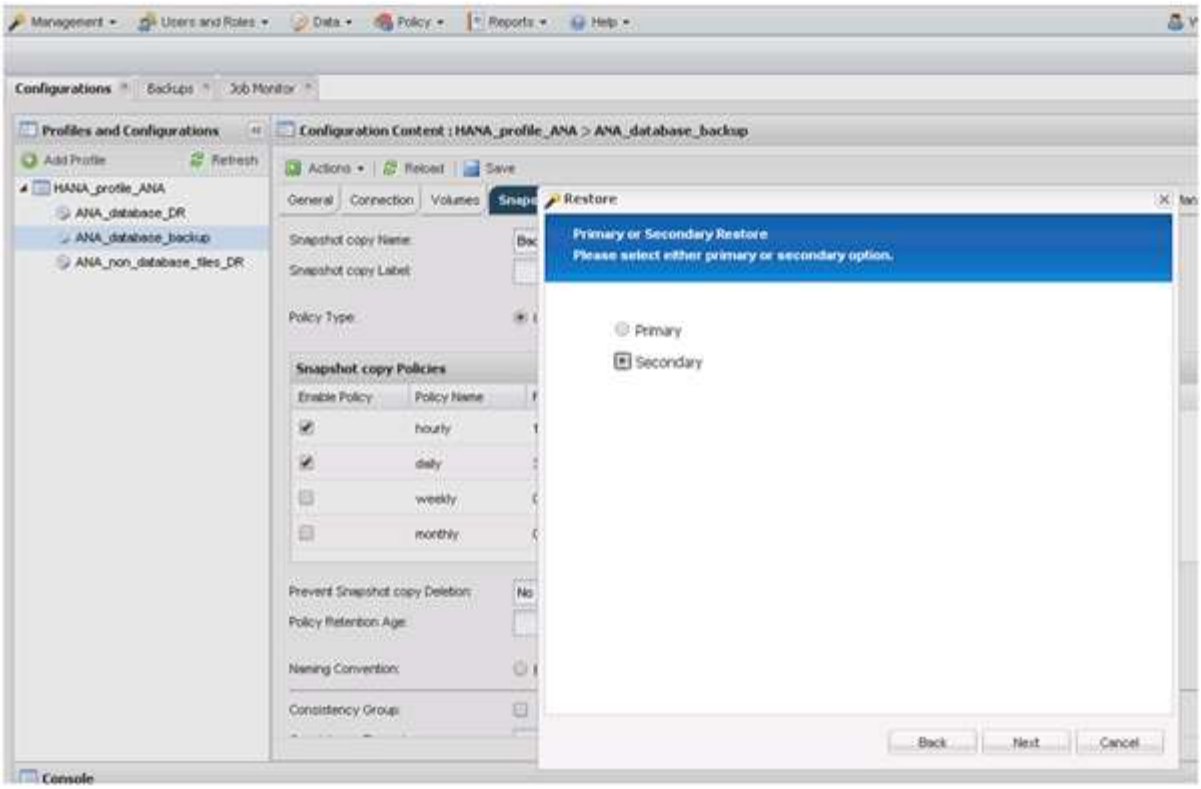
5. 移至SnapCreator GUI。
6. 選取SAP HANA系統、然後按一下「動作>*還原*」。



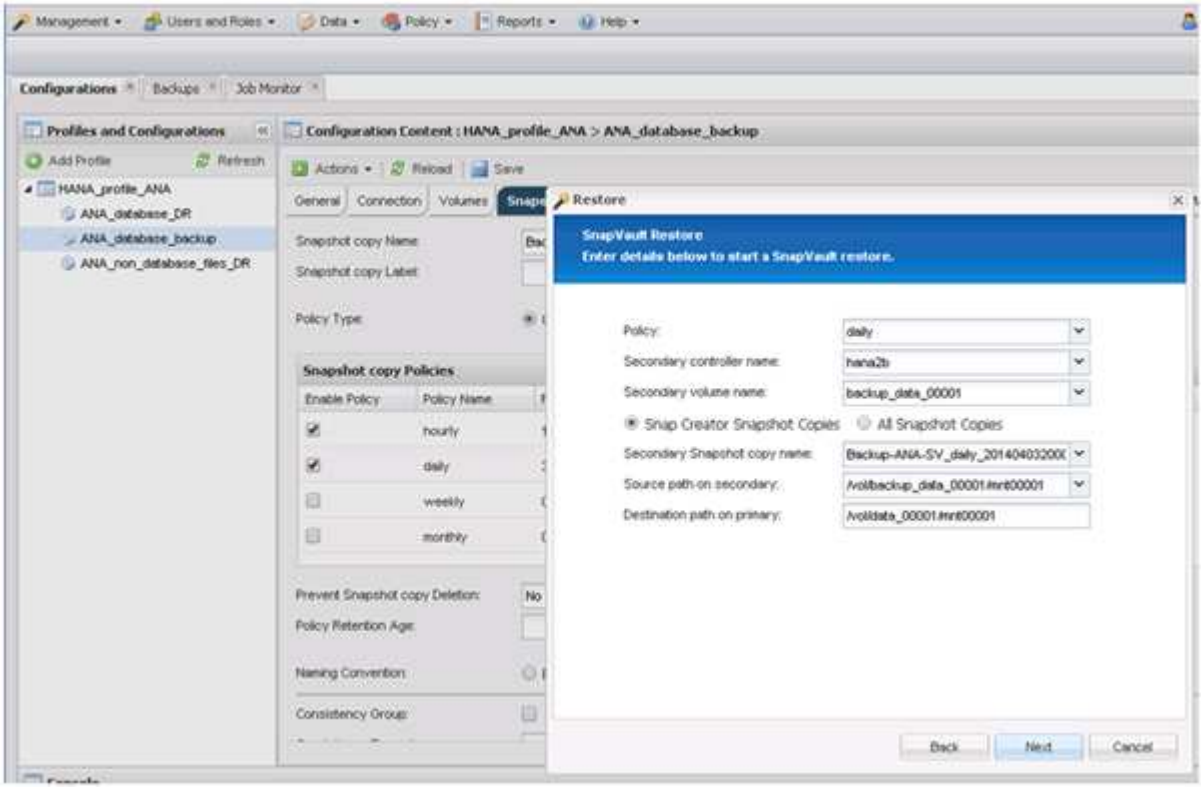
歡迎畫面隨即出現。



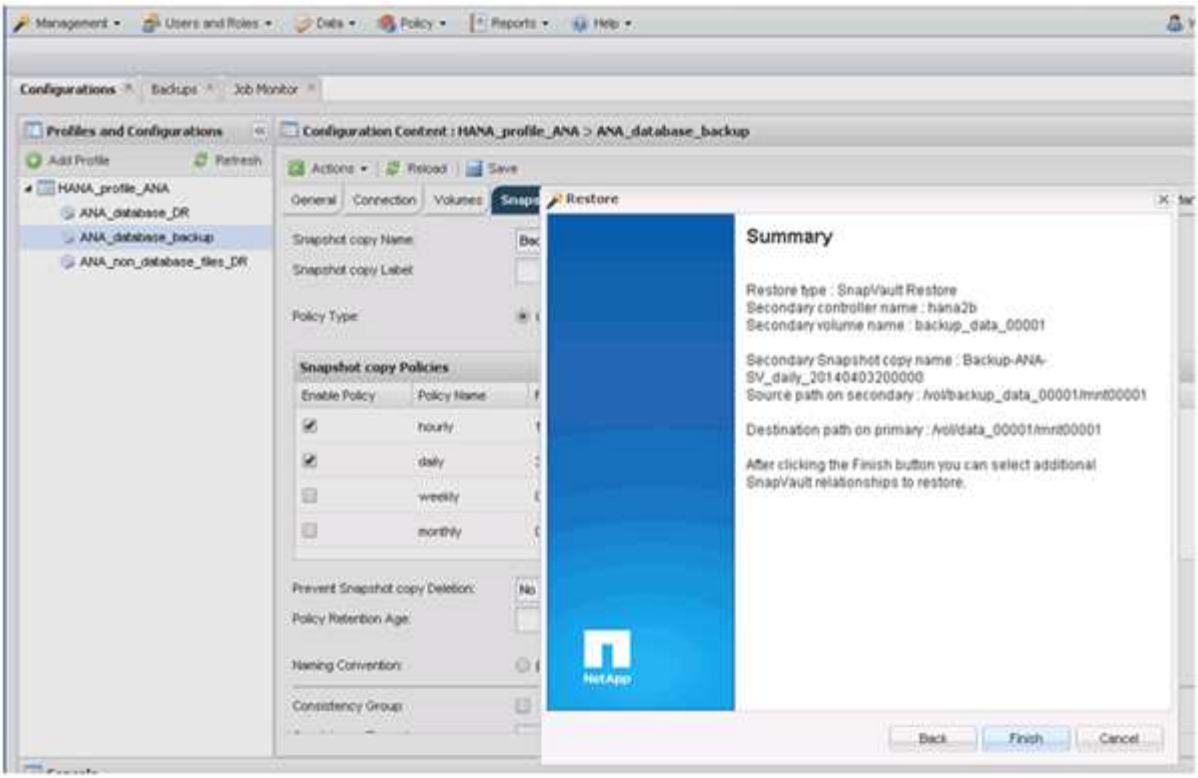
7. 單擊 * 下一步 * 。
8. 選擇*二級*並按*下一步*。



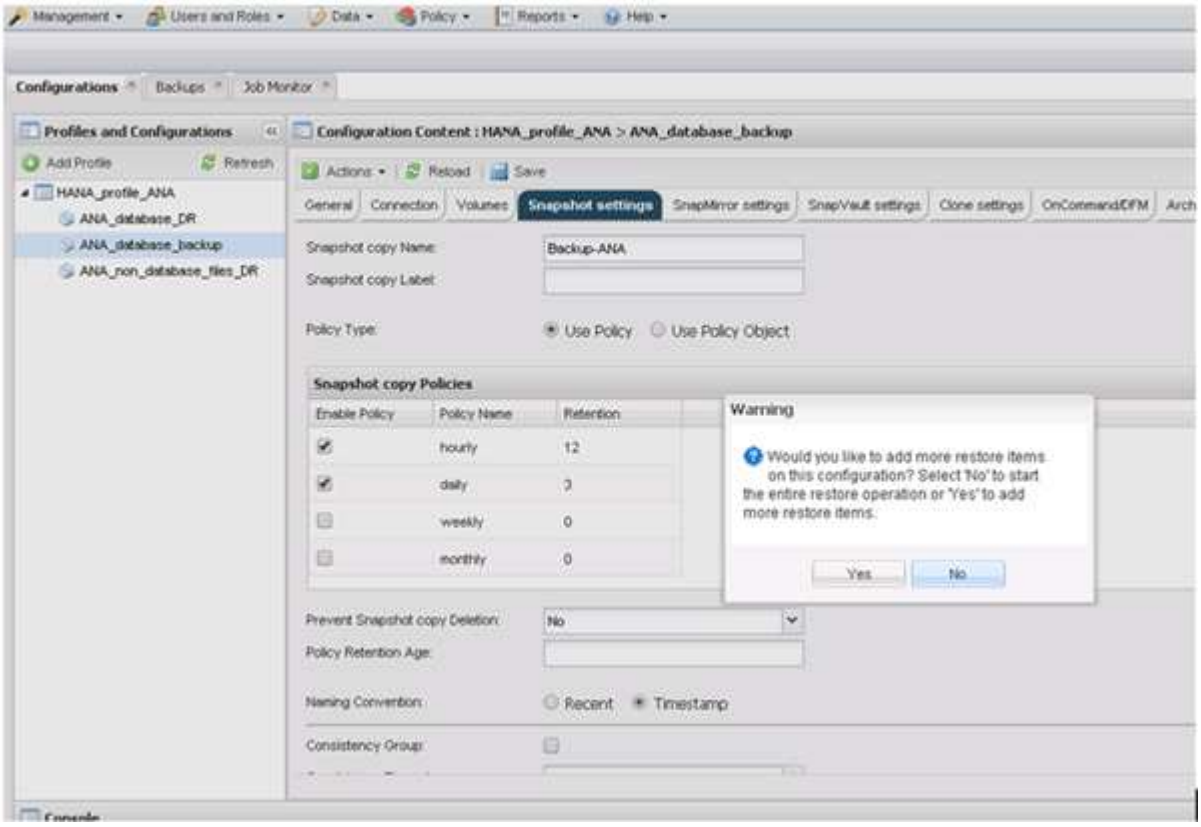
9. 輸入所需資訊。Snapshot名稱與SAP HANA Studio中選取的備份ID有關。



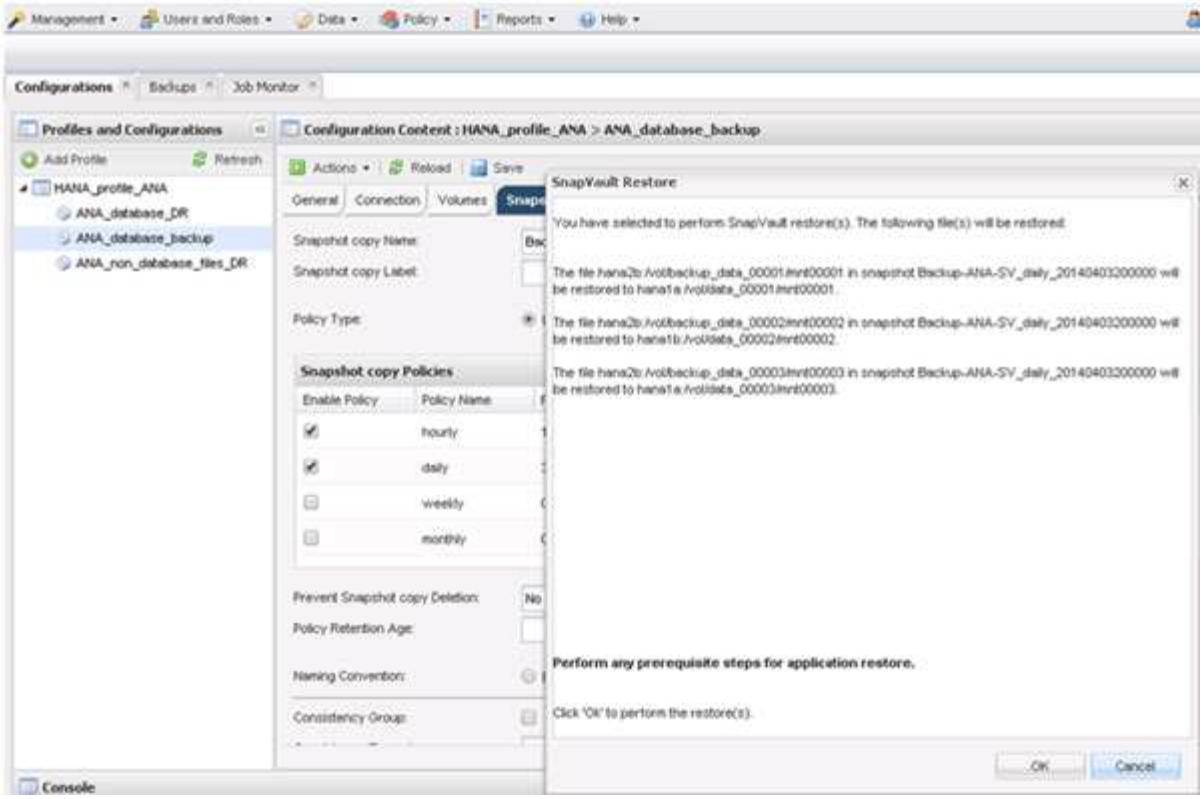
10. 選擇*完成*。



11. 按一下「是」以新增更多要還原的項目。



12. 針對所有需要還原的磁碟區、提供必要資訊。在設定data_00001、data_00002和data_00003中、需要選取以進行還原程序。



13. 選取所有磁碟區後、選取*確定*以開始還原程序。

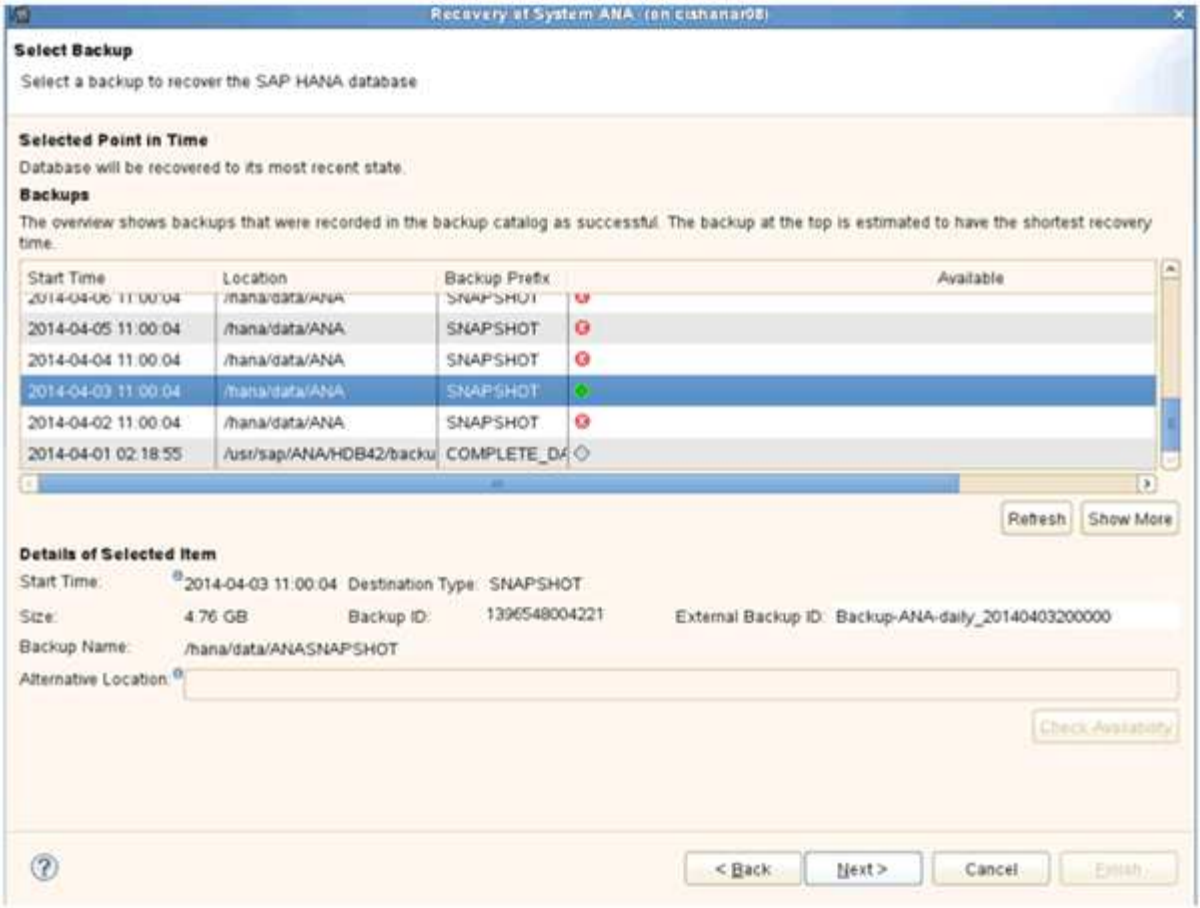
    等待還原程序完成。

14. 在每個資料庫節點上、重新掛載所有資料磁碟區、以清除「『NFS句點』」。

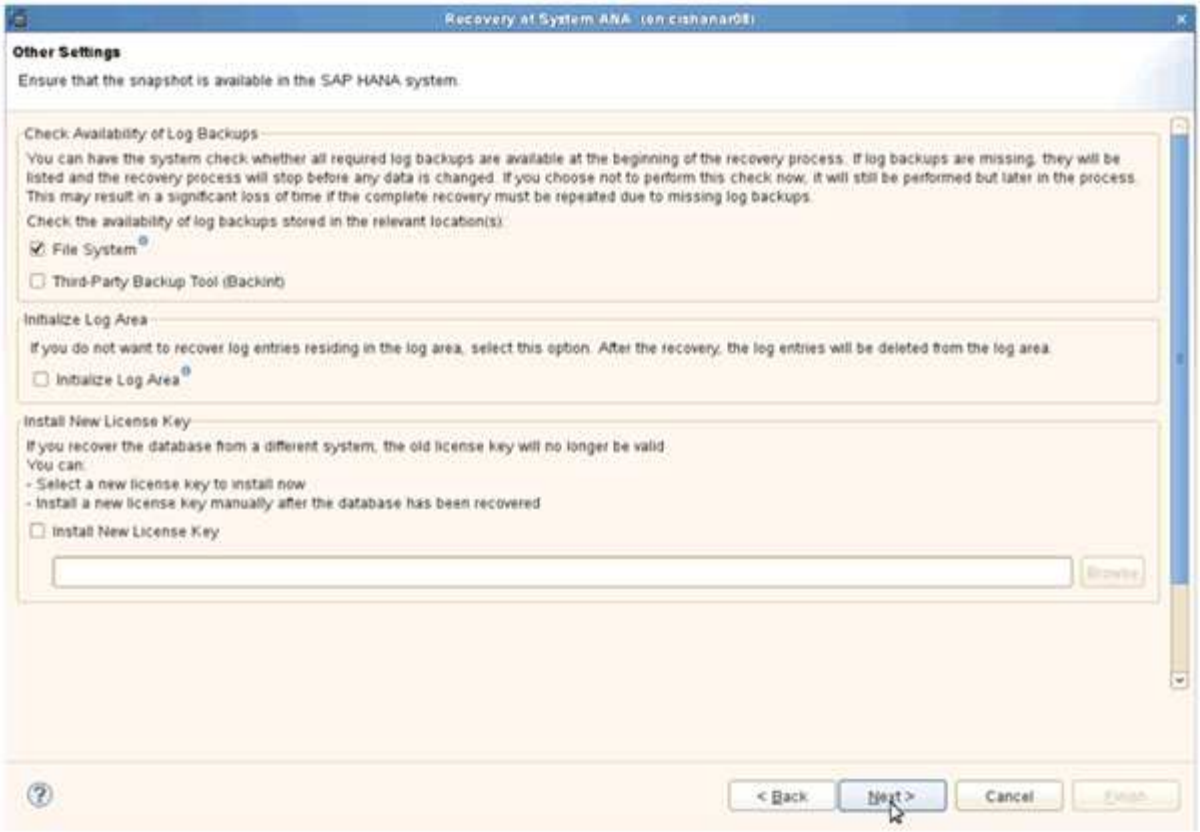
    在此範例中、所有三個磁碟區都需要在每個資料庫節點上重新掛載。

```
mount -o remount /hana/data/ANA/mnt00001
mount -o remount /hana/data/ANA/mnt00002
mount -o remount /hana/data/ANA/mnt00003
```

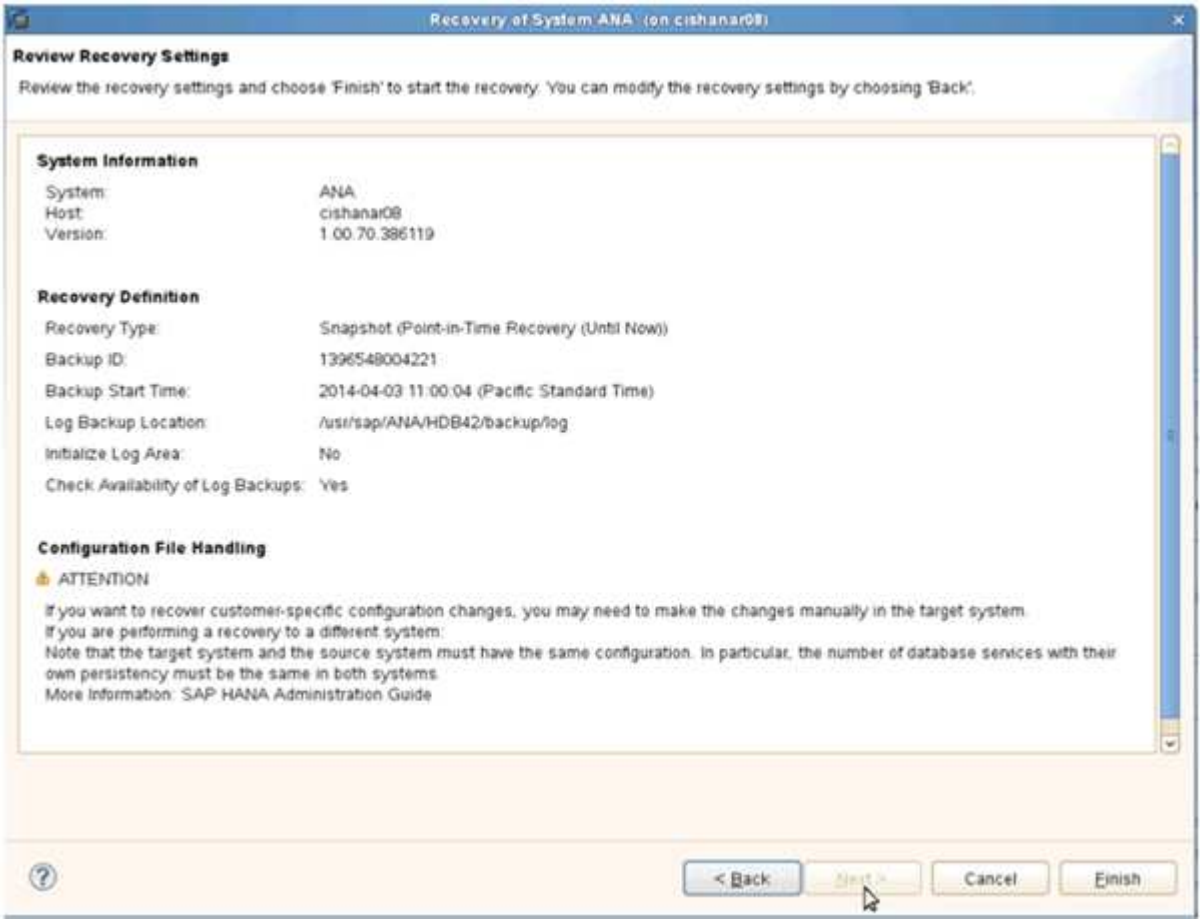
15. 前往SAP HANA Studio、然後按一下「* Refresh*（重新整理）」以更新備份清單。



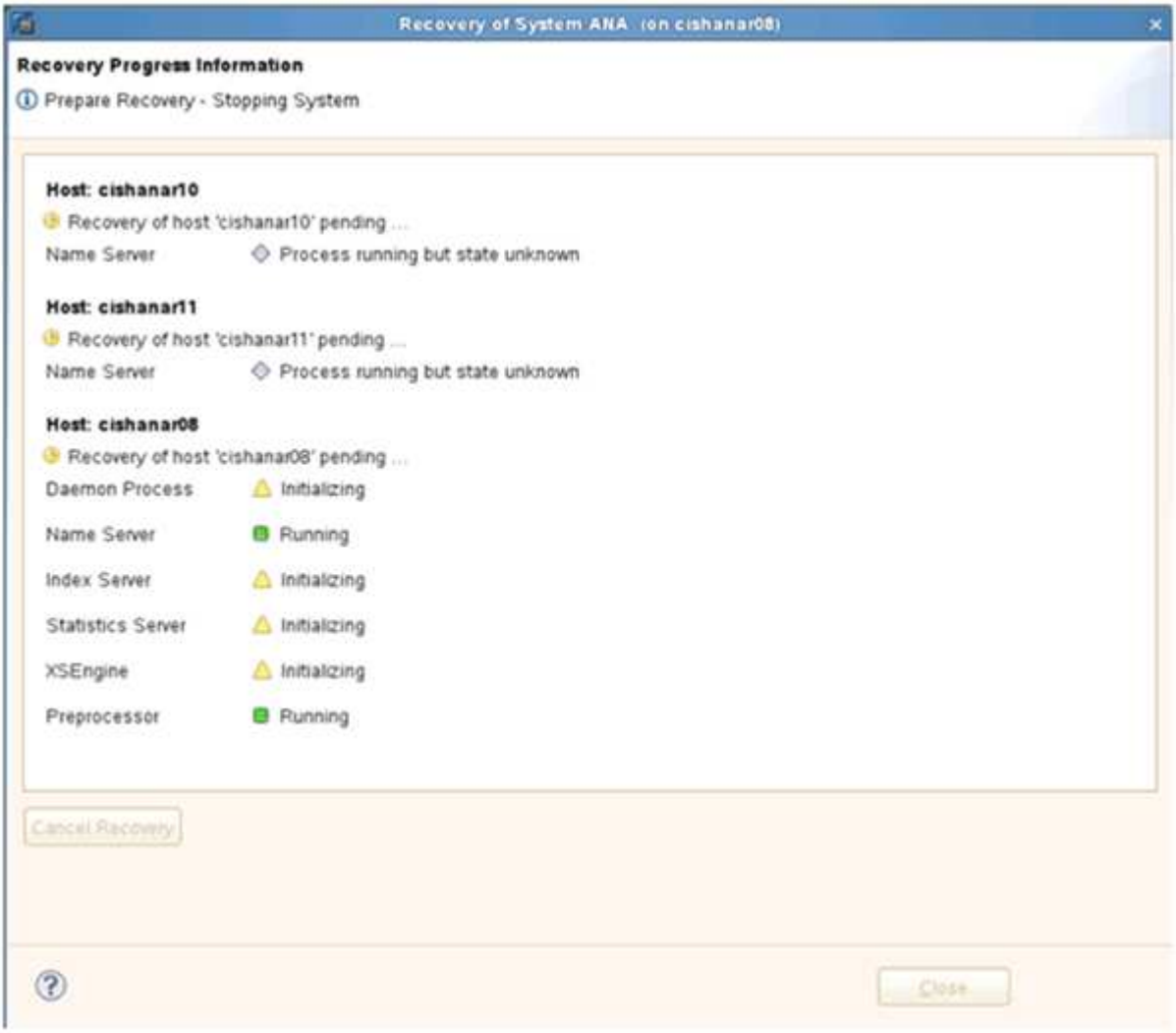
16. 使用Snap Creator還原的備份會在備份清單中顯示綠色圖示。選取備份、然後按一下*下一步*。
17. 視需要選取其他設定、然後按一下*下一步*。


Recovery of System ANA (on cishanar08)
Other Settings
Ensure that the snapshot is available in the SAP HANA system.
Check Availability of Log Backups
You can have the system check whether all required log backups are available at the beginning of the recovery process. If log backups are missing, they will be listed and the recovery process will stop before any data is changed. If you choose not to perform this check now, it will still be performed but later in the process. This may result in a significant loss of time if the complete recovery must be repeated due to missing log backups.
Check the availability of log backups stored in the relevant location(s):
File System
Third-Party Backup Tool (Backint)
Initialize Log Area
If you do not want to recover log entries residing in the log area, select this option. After the recovery, the log entries will be deleted from the log area.
Initialize Log Area
Install New License Key
If you recover the database from a different system, the old license key will no longer be valid.
You can:
- Select a new license key to install now
- Install a new license key manually after the database has been recovered
Install New License Key
Browse...
< Back
Next >
Cancel
Finish

18. 單擊*完成*。


Recovery of System ANA (on cishanar08)
Review Recovery Settings
Review the recovery settings and choose 'Finish' to start the recovery. You can modify the recovery settings by choosing 'Back'.
System Information
System: ANA
Host: cishanar08
Version: 1.00.70.386119
Recovery Definition
Recovery Type: Snapshot (Point-in-Time Recovery (Until Now))
Backup ID: 1396548004221
Backup Start Time: 2014-04-03 11:00:04 (Pacific Standard Time)
Log Backup Location: /usr/sap/ANA/HDB42/backup/log
Initialize Log Area: No
Check Availability of Log Backups: Yes
Configuration File Handling
ATTENTION
If you want to recover customer-specific configuration changes, you may need to make the changes manually in the target system.
If you are performing a recovery to a different system:
Note that the target system and the source system must have the same configuration. In particular, the number of database services with their own persistency must be the same in both systems.
More Information: SAP HANA Administration Guide
< Back
Next >
Cancel
Finish

恢復程序隨即開始。



19. 恢復程序完成後、SnapVault 如有需要、請繼續執行「還原關係」。

還原後恢復SnapVault 不還原關係

任何未使用最新Snapshot備份完成的還原、都會刪除SnapVault 主儲存系統的還原關係。

還原與還原程序完成後、SnapVault 必須恢復不二關係、才能使用Snap Creator再次執行備份。否則、Snap Creator會發出錯誤訊息、因為SnapVault 它無法在主要儲存系統上找到更多的相關資訊。

如果來源磁碟區與目的地磁碟區之間仍有通用的Snapshot複本、則所需的資料傳輸將以差異傳輸為基礎。

恢復**SnapVault** 以**Data ONTAP 7-Mode**運作的支援功能與支援功能之間的不正常關係

如果您使用非最新的Snapshot備份來還原、則需要恢復SnapVault 「支援」關係、以便Snap Creator繼續執行備份。

1. 輸入下列命令、以SnapVault 7-Mode繼續執行支援的功能、恢復執行支援的功能。Data ONTAPSetsstart -r -S sSOUR_控制 器：source_volumeBackup控制器：SnapVault backup_volume

   針對屬於SAP HANA資料庫的所有磁碟區執行此步驟。

```
hana2b> snapvault start -r -S hana1a:/vol/data_00001/mnt00001
hana2b:/vol/backup_data_00001/mnt00001
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:08:21 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00001/mnt00001 to hana1a:/vol/data_00001/mnt00001 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana2b> snapvault start -r -S hana1b:/vol/data_00002/mnt00002
hana2b:/vol/backup_data_00002/mnt00002
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:09:49 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00002/mnt00002 to hana1b:/vol/data_00002/mnt00002 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana2b> snapvault start -r -S hana1a:/vol/data_00003/mnt00003
hana2b:/vol/backup_data_00003/mnt00003
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:10:25 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00003/mnt00003 to hana1a:/vol/data_00003/mnt00003 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

資料傳輸完成後、您可以使用Snap Creator重新排程備份。

使用**SnapVault** 叢集**Data ONTAP** 式的不一樣功能恢復不一樣的關係

如果您使用非最新的Snapshot備份來還原、則需要恢復SnapVault 「支援」關係、以便Snap Creator繼續執行備份。

1. 重新建立SnapVault 及重新同步化此等資訊關係。

```
hana::> snapmirror create -source-path hana1a:hana_data -destination
-path
hana2b:backup_hana_data -type XDP
Operation succeeded: snapmirror create the relationship with destination
hana2b:backup_hana_data.

hana::> snapmirror resync -destination-path hana2b:backup_hana_data
-type XDP

Warning: All data newer than Snapshot copy sc-backup-
daily_20140430121000 on volume
hana2b:backup_hana_data will be deleted.
Do you want to continue? {y|n}: y
[Job 6554] Job is queued: initiate snapmirror resync to destination
"hana2b:backup_hana_data".
[Job 6554] Job succeeded: SnapMirror Resync Transfer Queued
```

2. 若要實際重新啟動SnapVault 此功能、需要手動複製Snapshot複本。

```
hana::> snapshot create -vserver hana1a -volume hana_data -snapshot
sv_resync

hana::> snapshot modify -vserver hana1a -volume hana_data -snapshot
sv_resync -snapmirror-label daily

hana::> snapmirror update -destination-path hana2b:backup_hana_data
Operation is queued: snapmirror update of destination
hana2b:backup_hana_data.
```

3. 驗證SnapVault 目標清單中是否顯示了此等不景關係。

```
hana::> snapmirror list-destinations -source-path hana1a:hana_data
                                                      Progress
Source             Destination         Transfer   Last
Relationship
Path         Type  Path         Status Progress   Updated      Id
----------- ----- ------------ ------- --------- ------------
---------------
hana1a:hana_data
            XDP   hana2b:backup_hana_data
                                Transferring
                                        38.46KB   04/30 18:15:54
                                                                 9137fb83-
cba9-11e3-85d7-123478563412
```

在主儲存設備故障後還原資料庫

主儲存設備故障、或從主儲存設備的磁碟區刪除所有Snapshot複本之後、Snap Creator將無法處理還原、因為SnapVault 在主儲存系統上不再存在不可用的情況。

在主儲存設備故障後還原資料庫、**Data ONTAP** 以**7-Mode**運作

在Data ONTAP 執行以7-Mode運作的主儲存系統故障後、您可以還原SAP HANA資料庫。

1. 在這種情況下、必須使用下列命令直接在次要儲存系統上執行還原SnapVault ：-s snapshot _name -S backup_controller：backup_volumesource_controller：source_volume

   針對屬於SAP HANA資料庫的所有磁碟區執行此步驟。

```
hana1a>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00001/mnt00001 hana1a:/vol/data_00001/mnt00001
Restore will overwrite existing data in /vol/data_00001/mnt00001.
Are you sure you want to continue? y
Thu Apr 10 11:55:55 CEST [hana1a:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00001/mnt00001.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana1a>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00003/mnt00003 hana1a:/vol/data_00003/mnt00003
Restore will overwrite existing data in /vol/data_00003/mnt00003.
Are you sure you want to continue? y
Thu Apr 10 11:58:18 CEST [hana1a:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00003/mnt00003.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana1b>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00002/mnt00002 hana1b:/vol/data_00002/mnt00002
Restore will overwrite existing data in /vol/data_00002/mnt00002.
Are you sure you want to continue? y
Thu Apr 10 12:01:29 CEST [hana1b:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00002/mnt00002.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

還原程序完成後、您可以使用SAP HANA來執行還原。

使用叢集**Data ONTAP** 式的功能來還原資料庫、因為主要儲存設備故障

執行叢集Data ONTAP 式功能的主要儲存系統故障後、您可以還原SAP HANA資料庫。

假設主要磁碟區已完全遺失、您需要建立新的主要磁碟區、然後從備份磁碟區還原。

1. 建立具有資料類型保護的主要磁碟區。

```
hana::> volume create -vserver hana1a -volume hana_data -aggregate
aggr_sas_101 -size 300G -state online -type DP -policy default -autosize
-mode grow_shrink -space-guarantee none
-snapshot-policy none -foreground true
[Job 6744] Job is queued: Create hana_data.
[Job 6744] Job succeeded: Successful
```

2. 從備份磁碟區還原所有資料。

```
hana::> snapmirror restore -destination-path hana1a:hana_data -source
-path hana2b:backup_hana_data -source-snapshot sc-backup-
daily_20140505121000
[Job 6746] Job is queued: snapmirror restore from source
"hana2b:backup_hana_data" for the
snapshot sc-backup-daily_20140505121000.

hana::> job show -id 6746
Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
6746   SnapMirror restore   hana        hana01        Running
       Description: snapmirror restore from source
"hana2b:backup_hana_data" for the snapshot sc-backup-
daily_20140505121000
```

還原程序完成後、您可以使用SAP HANA來執行還原。

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